CPTG Geometric Pi Branch

Cosmological Comparison Coordinates

Version III

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Abstract

This article is the canonical technical reference for the Curvature Polarization Transport Gravity (CPTG) geometric pi branch and its comparison-coordinate system. CPTG contains one native cosmological branch, the geometric π branch. The acoustic and luminosity branches are projections of that native branch into their respective observational coordinates; they are not independent cosmologies or additional native expansion laws. At the luminosity comparison layer, the same one-ninth increment is obtained through three algebraically distinct CPTG constructions: the frame-conditional matched polarization carrier response $\mathsf{C}_{\Pi,\star} = I/9$; the structural coefficient ratio $C_H/C_R = (16\pi/3)/(48\pi) = 1/9$; and the acoustic rank quotient $\mathcal{R}_3 \equiv \Omega_{m,\mathrm{ac}}/\mathcal{G}_T = [p_{\mathrm{ac}}/(3\pi)]/[p_{\mathrm{ac}}/\pi] = 1/3$, whose square is $1/9$. These internal derivational routes converge on the registered affine comparison map $\mathsf{Q}_L^{\mathrm{aff}} \equiv I + \mathsf{C}_\Pi$, giving $\mathcal{G}_L^{\mathrm{aff}} = 10/9$, $H_L^{\mathrm{aff}}(a) = \sqrt{10/9}\, H_\pi(a)$, and therefore $E_L^{\mathrm{aff}}(a) = E_\pi(a)$ exactly. No SH0ES quantity enters these theory-side derivations; the four-anchor SH0ES result is used afterward as an independent external observational validation of the derived endpoint. The reference also records the interface between redshift-selected native phase and object-dependent bound-system structure, and introduces the symmetric acoustic–luminosity midpoint prescription $\omega_{\mathrm{AL}} = 1/2$ for calculations that require a central bracket between those two projected coordinates.

For CMB map and spectral comparison, SkyPi supplies the current authoritative construction; the earlier Planck/WMAP map-closure operator is retained as a legacy implementation record.

The purpose of this guide is operational rather than evidentiary: it records the quantities, dependency chains, registry classes, transport maps, fixed inputs, derived values, implementation conventions, and sector-specific coordinate constructions used across CPTG cosmology, bound-system, and nuclear-reaction papers. The common methods include a data-role firewall that forbids downstream validation information from flowing back into the solve, the post-solution Structural Mode N readout, a hierarchy of observation operators matched to the support geometry and observable order, and the exact four-sector nuclear incidence/rank-four source construction. A reader should be able to begin with the native branch, identify the correct upstream quantities for a target sector, apply the declared CPTG map in the required order, distinguish derived quantities from frozen inputs or data-conditioned coordinates, and locate the corresponding implementation and provenance record.

The guide is organized in the order in which the framework is used: native branch and scale architecture first; the cross-scale bound-system/native-phase interface second; quantity taxonomy and parameter registry third; then the nuclear/primordial, CMB/acoustic, spectral-response, growth/lensing, distance, common-phase clock and horizon, BAO, and ShapeFit construction layers; followed by implementation checks, validation records, open conditions, and evidence identities. The validation sections document how the registered quantities have been exercised and where their current implementation boundaries lie; they are not used here as a proof of CPTG. Dedicated CPTG papers remain the sources for sector-specific derivations, tests, and scientific arguments, while this document supplies the common technical vocabulary and reference equations needed to keep those papers mutually consistent.

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1 How to Use This Reference

This document is intended to be used as a technical lookup and construction guide. It is the common reference layer for CPTG papers that need the geometric pi branch, its derived quantities, or one of its observational or network comparison coordinates.

CPTG is treated throughout as a locked, scalable framework. Native geometric quantities, structural laws, and transport operators are fixed before observational evaluation. Scale, phase, spectral, or network transport therefore means deterministic propagation of a fixed geometric law into another scale or comparison coordinate; it does not permit observational data to feed backward and retune the native geometry or transport rule. Conventional external cosmological solver machinery, where used, belongs only to the comparison/translation layer and is not promoted into CPTG-native dynamics.

The recommended workflow is:

1. identify the physical sector or reported observable to be constructed;
2. begin from the native branch quantities and the scale/epoch architecture defined below;
3. for a bound system, keep the redshift-selected native quantities separate from the object's locally inferred structural quantities before forming any cross-scale comparison;
4. use the parameter registry to determine the status of every required quantity before using it;
5. follow the sector-specific dependency chain in the order stated, without skipping directly from an observational value to a native CPTG quantity;
6. keep frozen comparison inputs and fixed implementation settings distinct from quantities derived from the CPTG branch;
7. apply the no-retuning rule universally; diagnostic protocols may vary declared downstream controls or stress coordinates but may not alter registered CPTG quantities after target exposure; and
8. consult the later implementation, validation, and evidence sections only to determine how the construction has been exercised, what numerical authority was used, and what limitations remain open.

1.1 Data-Role Firewall

The locked-framework rule is enforced by a data-role firewall. Every quantity entering a CPTG calculation must have an upstream role before the calculation begins, and no downstream comparison result is permitted to migrate backward into the theory or structural solve. The universal inference direction is

$$\begin{gathered}\text{declared physical and observational inputs}\\ \downarrow\\ \text{fixed CPTG solve or registered transport map}\\ \downarrow\\ \text{derived CPTG coordinates and structural readouts}\\ \downarrow\\ \text{declared downstream observational or network comparison.}\end{gathered} \tag{1}$$

There is no reverse arrow. A downstream observable may falsify, qualify, or support the declared construction, but it does not change a native invariant, structural law, transport operator, registry threshold, source coefficient, or object-level solution after exposure. The label "held-out" is reserved for comparison lanes that were actually frozen before target exposure; externally anchored, data-conditioned, and diagnostic lanes retain their own declared status. Inferring an

object-dependent structural state such as $a_\star$ from declared object data under a fixed CPTG law is structural inference; changing a registered law, coefficient, gate, or transport rule after comparison exposure is retuning.

The same rule applies to observational geometry. Quantities compared in one closure statement must refer to the same declared aperture, support region, epoch coordinate, and source convention unless an explicit conversion operator is applied first. A radius from one aperture, a mass from another, and a support measure from a third are not interchangeable inputs merely because they describe the same object. This same-aperture/same-structure discipline is therefore enforced directly as an operational rule of this reference.

Role	Allowed use	Firewall boundary
Declared solve input	Enters the fixed field, structural, or transport calculation	Must be identified before the target comparison is opened
Fixed implementation setting	Specifies the declared numerical or adapter protocol	Held fixed during the calculation; not promoted to a new theory degree of freedom
Derived CPTG quantity	Computed from the fixed solve or registered map	May flow downstream only; cannot be overwritten by a validation residual
Validation-only observable	Tests the retained CPTG result	Cannot select, tune, or repair an upstream CPTG quantity
Diagnostic control	Robustness and representation tests	May diagnose implementation sensitivity but does not redefine the registered construction

Table 1: Universal CPTG data-role firewall. Sector-specific papers may refine these roles, but they do not reverse the inference direction in Equation (1).

The equations and registry entries in the construction sections are the reference layer. Validation statistics, likelihood outputs, network residuals, and diagnostic scans document implementation behavior; they do not redefine the registered CPTG quantities. When a sector-specific CPTG paper gives a more detailed derivation or evidence analysis, that paper supplies the specialized argument while this guide supplies the common definitions and coordinate relationships.

2 Core Branch and Reference Conventions

CPTG uses one native geometric π branch. The acoustic and luminosity branches are declared projections into observational coordinates, not additional native cosmologies; comparison coordinates express the fixed theory in sector-specific variables.

The cosmological branch architecture is therefore

$$\mathfrak{B}_\pi \xrightarrow{\mathcal{P}_{\rm ac}} \mathfrak{B}_{\rm ac}, \qquad \mathfrak{B}_\pi \xrightarrow{\mathcal{P}_{\rm lum}} \mathfrak{B}_{\rm lum}, \tag{2}$$

where $\mathfrak{B}_\pi$ is native and $\mathfrak{B}_{\rm ac}$ and $\mathfrak{B}_{\rm lum}$ are projected branches sharing the native phase coordinate

$$N_\pi \equiv \ln a, \tag{3}$$

The subscript distinguishes cosmological phase N_π from post-solution Structural Mode N. For any branch b with defined rate history $H_b(a)$,

$$H_b\, dt_b = dN_\pi, \qquad t_b(a) - t_b(a_{\rm ref}) = \int_{a_{\rm ref}}^{a} \frac{da'}{a' H_b(a')}. \tag{4}$$

Branch synchronization is internal; no common clock or cross-branch thermal synchronization is imposed. The accumulated separation from native time is

$$\Delta t_{b\pi}(a) = \int_{a_{\rm ref}}^{a} \frac{da'}{a'} \left[\frac{1}{H_b(a')} - \frac{1}{H_\pi(a')} \right], \tag{5}$$

The native thermal property

$$\Theta_{\pi T}(a) \equiv \frac{T(a)}{T_0} = 1 + z = a^{-1}, \qquad T_0 = 2.72548 \text{ K}, \tag{6}$$

belongs to $\mathfrak{B}_\pi$; it is not an additional cosmological branch [1].

The comparison hierarchy is

$$\begin{gathered}\text{CPTG-native geometric quantity}\\ \downarrow\\ \text{comparison-coordinate quantity}\\ \downarrow\\ \text{reported observational summary}\end{gathered} \tag{7}$$

The hierarchy separates fixed geometry, sector-specific comparison coordinates, and the reported observational summary.

The native and projected branch values used here are

$$p_C = \pi, \tag{8}$$

$$p_{\mathrm{ac}} \equiv 3 - \frac{\pi}{100} \simeq 2.968584073464. \tag{9}$$

The displayed twelve-decimal value is the locked numerical representation used by the comparison pipelines; the analytic definition remains $3 - \pi/100$.

$$\mathcal{G}_T \equiv \frac{p_{\mathrm{ac}}}{p_C} \simeq 0.9449296585513721, \tag{10}$$

with rate factor

$$\sqrt{\mathcal{G}_T} \simeq 0.9720749243506758. \tag{11}$$

The native late and transport weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \qquad \mathcal{T}_\pi = \frac{1}{\pi}, \tag{12}$$

so the native background is $E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}$.

The acoustic projection is fixed at the background level by subtracting the native late term,

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}, \tag{13}$$

and applying the exact normalized projector

$$\begin{aligned} X_{\mathrm{ac}}(a) &\equiv \mathcal{T}_\pi^{1-\mathcal{G}_T} X_\pi(a)^{\mathcal{G}_T} \\ &= \mathcal{T}_\pi a^{-p_{\mathrm{ac}}}, \end{aligned} \tag{14}$$

$$E_{\mathrm{ac}}(a)^2 = \mathcal{B}_\pi + X_{\mathrm{ac}}(a). \tag{15}$$

The prefactor preserves $X_{\mathrm{ac}}(1) = X_\pi(1) = \mathcal{T}_\pi$, so the exponent change remains a projection of the native transport term.

The native geometric acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} = 5.568327996831708 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{16}$$

The native geometric normalization is

$$H_0^{(\pi)} = 69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{17}$$

and the acoustic comparison normalization, also used by the CMB coordinate, is

$$H_{0,\mathrm{ac}}^{\mathrm{CPTG}} \equiv H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{18}$$

For the luminosity projection, define the acoustic comparison weight

$$\Omega_{m,\mathrm{ac}} \equiv \frac{p_{\mathrm{ac}}}{3\pi}. \tag{19}$$

With $\mathcal{G}_T = p_{\mathrm{ac}}/\pi$, the acoustic branch supplies the exact rank quotient

$$\mathcal{R}_3 \equiv \frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T} = \frac{p_{\mathrm{ac}}/(3\pi)}{p_{\mathrm{ac}}/\pi} = \frac{1}{3}. \tag{20}$$

The later matched-response and structural-coefficient derivations close on the same one-ninth increment, giving

$$\mathcal{G}_L^{\mathrm{aff}} = 1 + \mathcal{R}_3^2 = 1 + \left(\frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T}\right)^2 = \frac{10}{9}, \tag{21}$$

with rate and reciprocal clock factors

$$\Gamma_L^{\mathrm{aff}} = \sqrt{\frac{10}{9}}, \qquad \Xi_L^{\mathrm{aff}} = \sqrt{\frac{9}{10}}. \tag{22}$$

The matched-normalization luminosity projection acts on the native branch as

$$H_L^{\mathrm{aff}}(a) = \Gamma_L^{\mathrm{aff}} H_\pi(a) = \sqrt{\frac{10}{9}}\, H_\pi(a), \tag{23}$$

with exactly native normalized shape,

$$E_L^{\mathrm{aff}}(a) \equiv \frac{H_L^{\mathrm{aff}}(a)}{H_{0,L}^{\mathrm{aff}}} = E_\pi(a), \tag{24}$$

and branch clock

$$\frac{dt_L^{\mathrm{aff}}}{dt_\pi} = \Xi_L^{\mathrm{aff}} = \sqrt{\frac{9}{10}}, \qquad H_L^{\mathrm{aff}} dt_L^{\mathrm{aff}} = H_\pi dt_\pi = dN_\pi. \tag{25}$$

At the normalized present phase this gives the theory-side luminosity endpoint

$$H_{0,L}^{\mathrm{aff}} = H_0^{(\pi)}\sqrt{\frac{10}{9}} = 73.1711530416\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{26}$$

No SH0ES quantity enters these relations; the distance ladder is used later only as independent external validation. Section 11.2 gives the full polarization-response, structural-coefficient, and acoustic-rank derivations.

The framework and public computational materials are documented in Refs. [38, 51].

3 Scale and Native-Phase Architecture

CPTG is a framework of nested physical scales. The presentation therefore uses physical scale as its primary organizational axis, proceeding from microscopic nuclear structure to cosmological distance scales. Within each scale regime, the secondary ordering follows native geometric phase, equivalently the observed redshift coordinate, from the earliest validated phase toward the normalized present. Branch-specific times are obtained only after that phase has been selected and are not used as the ordering variable. The phrase "earliest phase" is used deliberately: the comparison program begins with the earliest phase for which the present paper supplies a defined and tested coordinate, not with a claim about the mathematical limit $t = 0$.

Regime	Physical scale	Epoch ordering	Comparison layers
I	Scale-independent geometric foundation and cross-scale interfaces	Applies across all validated native phases	Native pi branch, bound-system/native-phase interface, registry classes, and no-retuning comparison rule
II	Microscopic nuclear to primordial-network scale	Primordial BBN epoch	Reaction source and observable projection, native-network response, baryon/radiation transport, and live mass-seven abundance transport
III	Recombination and acoustic-horizon scale	Recombination and CMB observation	Acoustic transport, amplitude/visibility gates, spectral-response bridge, SkyPi source/map operators (legacy map-envelope test), CMB likelihood closure, and common-phase horizon containment
IV	Structure-growth to cosmological-distance scale	Post-recombination evolution to the present epoch	Growth and lensing, distance/ruler coordinates, DESI BAO/full-shape diagnostics, supernova distance shape, and clock/age comparison

Table 2: Scale–phase organization used in this article. Physical scale is the primary reading axis; native geometric phase is the secondary axis within each regime.

The resulting conceptual progression is

$$\begin{gathered}\text{microscopic reaction source}\\ \downarrow\\ \text{primordial network}\\ \downarrow\\ \text{recombination/acoustic sky}\\ \downarrow\\ \text{structure growth and standard rulers}\\ \downarrow\\ \text{present-epoch distance and clock comparisons.}\end{gathered} \tag{27}$$

Coordinates that bridge two regimes are placed at the transition where their physical interpretation is clearest. CMB lensing, for example, is sourced by the CMB but accumulates through intervening structure, so its compressed late-time amplitude is treated as a bridge from the acoustic regime into the structure-growth regime rather than as a separate fundamental scale.

4 Bound-System Interface to Native Phase

The cosmological branch identifies when an observation is represented on CPTG native geometry; it does not by itself determine the structural state of a bound object observed at that phase. This distinction is important whenever the comparison framework is applied to galaxies, clusters, or compact systems. Redshift selects the native phase N_π and therefore the native quantities $E_\pi(a)$, $a_\pi(a)$, and $R_C(a)$. The bound system supplies its own object-dependent structural acceleration $a_\star$ through the appropriate structural or dynamical reduction [38, 47].

For the positive structural branch,

$$R_c = \frac{48\pi c^2}{a_\star}, \qquad a_\pi(a) = a_{\pi,0} E_\pi(a), \qquad R_C(a) = \frac{48\pi c^2}{a_\pi(a)}. \tag{28}$$

Here lowercase R_c is the bound-system structural curvature scale and uppercase R_C is the native cosmological curvature scale. Neither is the visible radius of the gravitating object. At the continuous endpoint $a_\star = 0$, the structural scale satisfies $R_c \to \infty$ while $a_\star$ itself remains the regular variable.

Whenever $a_\pi > 0$, the cross-scale acceleration ratio has the continuous definition

$$\eta_\star \equiv \frac{a_\star}{a_\pi}. \tag{29}$$

For $a_\star > 0$, the common geometric numerator additionally gives

$$\eta_\star = \frac{R_C}{R_c}. \tag{30}$$

Thus $a_\star = 0$ implies $\eta_\star = 0$ by direct evaluation of the acceleration ratio, while the radius form is understood by the continuous limit $R_c \to \infty$. Native phase selection alone does not require $a_\star$ to follow a_π and does not rescale the local bound-system solution.

The structural acceleration can also be written as the associated CPTG structural-rate coordinate

$$H_{\text{struct}} \equiv \frac{16\pi}{3}\frac{a_\star}{c}, \tag{31}$$

while the native branch satisfies

$$H_\pi(a) = \frac{a_\pi(a)}{c\sqrt{\mathcal{B}_\pi}}. \tag{32}$$

Their dimensionless cross-scale rate ratio is therefore

$$\begin{aligned} \mathcal{K}_\star &\equiv \frac{H_{\text{struct}}}{H_\pi} \\ &= \frac{16\pi}{3}\sqrt{\mathcal{B}_\pi}\,\eta_\star \\ &= 13.8338192981\,\eta_\star. \end{aligned} \tag{33}$$

Accordingly, $\mathcal{K}_\star$ is a derived cross-scale rate coordinate and carries no independent degree of freedom beyond $\eta_\star$. At $a_\star = 0$, $H_{\text{struct}} = 0$ and $\mathcal{K}_\star = 0$ continuously.

Define the native transport fraction

$$f_\pi(a) \equiv \frac{\mathcal{T}_\pi a^{-\pi}}{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}. \tag{34}$$

Because

$$\frac{d\ln E_\pi}{dN_\pi} = -\frac{\pi}{2} f_\pi(a), \tag{35}$$

the positive-branch logarithmic state–phase relation is

$$\frac{d\ln\eta_\star}{dN_\pi} = \frac{d\ln a_\star}{dN_\pi} + \frac{\pi}{2} f_\pi(a), \qquad a_\star > 0. \tag{36}$$

The logarithmic form is not evaluated at $a_\star = 0$; the unnormalized quantities above provide the continuous endpoint.

Let $\mathcal{S}$ denote the intrinsic solved structural state of the bound system: the source distribution, geometry, support state, inferred $a_\star$, and corresponding solved local field. The recent NMBS-C7447 comparison isolates two distinct comparison operations based on Eq. (36) [45, 46, 47]. Under a strict fixed-local-state relocation,

$$\left.\frac{\partial a_\star}{\partial N_\pi}\right|_{\mathcal{S}} = 0, \tag{37}$$

so the source, $a_\star$, R_c, solved local field, and any post-solution Structural Mode remain fixed while the native quantities move with phase. Under a native-locked comparison trajectory,

$$\frac{d\ln\eta_\star}{dN_\pi} = 0, \tag{38}$$

which instead requires

$$\frac{d\ln a_\star}{dN_\pi} = -\frac{\pi}{2} f_\pi(a) = \frac{d\ln E_\pi}{dN_\pi}. \tag{39}$$

Because $\mathcal{K}_\star$ is proportional to $\eta_\star$, native locking also gives $d\ln\mathcal{K}_\star/dN_\pi = 0$, while fixed local structure lets both cross-scale ratios follow the native phase response. The two operations are therefore mathematically distinct. A native-locked comparison changes $a_\star$ and R_c and consequently requires the local CPTG field to be re-solved; a fixed-local-state comparison does not. Neither continuation is implied merely by assigning the observation a native phase.

Actual physical evolution is a third operation. In general one may write

$$a_\star = \mathcal{A}[\mathcal{S}], \qquad \mathcal{S} = \mathcal{S}(N_\pi), \tag{40}$$

so the structural acceleration changes only insofar as the evolving bound-system state changes. Native locking could describe a special physical trajectory if the solved dynamics enforce it, but it is not imposed by the native branch itself.

The same distinction applies to measurement interfaces. The quantity $a_\star$ is the object-dependent structural acceleration regardless of how it is inferred. A one-radius pressure-support expression such as $\sigma_\star^2/R_e$ is a compressed estimator when only integrated kinematics are available; it is not the universal definition of $a_\star$. When resolved multi-radius dynamics are available, a field-selected multi-radius reconstruction supersedes the one-radius estimate as the more informative measurement of the same structural quantity. The inferred cross-scale quantities $\eta_\star$ and $\mathcal{K}_\star$ therefore inherit the measurement interface used for $a_\star$: one-radius and resolved values should not be pooled as though they have equivalent information content, and resolved solutions must retain their declared support, anisotropy, and geometry assumptions [48]. Re-expression of a published galaxy radius or mass through acoustic or luminosity distance coordinates is likewise a coordinate-sensitivity calculation; it does not by itself evolve or redefine the object's intrinsic $a_\star$.

Quantity	Defining relation	Class	Role / boundary
$a_\star$	Object-dependent structural acceleration from the declared bound-system solve or estimator	Bound-system structural quantity	Same physical quantity across measurement interfaces; not set by redshift alone
R_c	$48\pi c^2/a_\star$ for $a_\star > 0$	Derived structural quantity	Bound-system curvature scale; $R_c \to \infty$ as $a_\star \to 0$
$\kappa_{\rm struct}$	1 kpc	Fixed structural readout setting	Registered radial normalization of the Structural Mode gradient term; not object-fitted
λ	Eq. (43)	Derived structural quantity	Continuous organization length of the solved field over the registered outer interval
N	R/λ	Derived structural quantity	Continuous post-solution Structural Mode; does not feed back into the field solve
$N_{\rm num}$	Eq. (48)	Derived structural quantity	Registered sampled Structural Mode readout; sampling protocol remains explicit
$\eta_\star$	$a_\star/a_\pi = R_C/R_c$ on the positive structural branch	Derived cross-scale coordinate	Acceleration/radius comparison; not phase invariant unless a continuation rule imposes it
$H_{\rm struct}$	$(16\pi/3)(a_\star/c)$	Derived structural quantity	Rate representation of the same structural scale
$\mathcal{K}_\star$	$H_{\rm struct}/H_\pi = (16\pi/3)\sqrt{\mathcal{B}_\pi}\,\eta_\star$	Derived cross-scale coordinate	Rate-space representation of $\eta_\star$; no independent degree of freedom

Table 3: Bound-system quantities and their interface to the native phase. The registry distinguishes the structural quantity itself from its derived cross-scale coordinates and from the observational method used to infer it.

4.1 Post-Solution Structural Mode N

Structural Mode N is a post-solution readout of where radial structural change is concentrated in a solved bound-system field; it does not enter the inference or field solve for $a_\star$ [49].

For a solved radial acceleration field $g(r)$, the registered reduced structural invariant is

$$C_g(r) = g^2(r) + \kappa_{\rm struct}^2 \left(\frac{dg}{dr}\right)^2, \qquad \kappa_{\rm struct} = 1\,\text{kpc}. \tag{41}$$

Here $\kappa_{\rm struct} = 1\,\text{kpc}$ is the fixed registered readout normalization, not an object-fitted quantity.

For outer readout radius R, define

$$w(r) = \left|\frac{dC_g}{dr}\right|. \tag{42}$$

The organization length is the weighted centroid over the registered outer interval,

$$\lambda = \frac{\displaystyle\int_{R/5}^{R} r\, w(r)\, dr}{\displaystyle\int_{R/5}^{R} w(r)\, dr}, \tag{43}$$

and the continuous Structural Mode is

$$N = \frac{R}{\lambda}. \tag{44}$$

Whenever $w(r)$ is nonzero on the registered interval,

$$1 \le N \le 5. \tag{45}$$

Smaller N places the structural centroid farther outward and larger N farther inward; CSMI is a downstream classification, not a solve input.

For sampled profiles, the registered realization sorts radii and applies the numerical derivative operator D_r defined in Ref. [49], then forms

$$C_i = g_i^2 + \kappa_{\mathrm{struct}}^2 (D_r g)_i^2, \qquad u_i = |(D_r C)_i| \,. \tag{46}$$

Retain only $r_i > R/5$; let n_{out} be the retained count and $\mathcal{S}_5$ the registered five-point filter of Ref. [49]. The sampled weights are

$$w_i = \begin{cases} (\mathcal{S}_5 u)_i, & n_{\mathrm{out}} > 5, \\ u_i, & n_{\mathrm{out}} \le 5. \end{cases} \tag{47}$$

The registered numerical readout is then

$$\lambda_{\mathrm{num}} = \frac{\sum_i r_i w_i}{\sum_i w_i}, \qquad N_{\mathrm{num}} = \frac{R}{\lambda_{\mathrm{num}}}. \tag{48}$$

The sampled estimator is not a quadrature rule for Equation (43); with nonnegative weights and nonzero denominator, $r_i > R/5$ gives

$$1 \le N_{\mathrm{num}} < 5. \tag{49}$$

Sampling-density controls belong to the dedicated Structural Mode implementation analysis [49].

The dependency chain is

$$\text{source and support} \longrightarrow a_\star \longrightarrow \text{solved } g(r) \longrightarrow C_g \longrightarrow w \longrightarrow \lambda \longrightarrow N \longrightarrow \text{CSMI}, \tag{50}$$

with no feedback from N or CSMI into $a_\star$ or the solved field.

4.2 Observation-Operator Hierarchy

Bound-system comparisons use the measurement operator appropriate to the system's geometry and support; the recurring hierarchy is

$$\begin{gathered} \text{one-radius / integrated support estimator} \\ \downarrow \\ \text{resolved second-moment tensor projection} \\ \downarrow \\ \text{higher-moment closure} \\ \downarrow \\ \text{distribution-function or orbit-superposition closure.} \end{gathered} \tag{51}$$

Each step asks a stronger observational question while retaining the upstream field; failure at a stronger level records failure or open closure there and does not retune the upstream law or retained structural state.

Resolved tensor and higher-moment implementations are documented in the Dragonfly 44 and XMM-VID1-2075 analyses [50, 48].

The hierarchy is an observation-interface rule: solve the field first, then apply the declared observation operator, preserving the firewall in Section 1.1.

5 Quantity Taxonomy and Parameter Registry

The registry separates native quantities, derived and structural quantities, comparison prescriptions and coordinates, frozen settings, pipeline results, and open diagnostics. Table 4 is the controlling definition of these classes and their permissible claim language; exact analytic projections remain derived quantities rather than additional native invariants.

Table 4: Claim-language rule associated with the paper-wide quantity taxonomy.

Registry class	Preferred claim language	Boundary
Native invariant	fixed	Geometric branch identity
Derived branch quantity	derived	Algebraic or integral consequence of native invariants
Parent geometric input	inherited / cited	Upstream CPTG result supplied by the named parent framework; not re-derived in this reference
Frame-conditional derived comparison response	derived within the named frame / completion / state	Exact response conditional on the declared structural frame and matched comparison state; no unique covariant frame-selection theorem implied
Bound-system structural quantity	inferred / selected by the declared structural solve	Object-specific structural quantity; measurement interface and support assumptions remain explicit
Derived structural quantity	derived from the named structural quantity	Algebraic structural consequence; not promoted to a native cosmological invariant
Fixed structural readout setting	fixed / registered by the readout protocol	Post-solution convention; not object-fitted and not a native cosmological invariant
Derived comparison coordinate	mapped / predicted in the declared coordinate	Arithmetic consequence of the declared comparison map; not a new native invariant
Registered comparison prescription	registered / applied	Fixed mapping rule; independent native/operator derivation is not implied unless cited
Transported comparison coordinate	transported from the named frozen anchor	Anchor provenance remains explicit
Bridge-validated comparison coordinate	bridge-validated	Native spectral-transport closure is not implied
Data-conditioned comparison coordinate	constructed / evaluated conditional on the declared scaffold	Observed or external scaffold is not promoted into CPTG theory
Abundance-transport comparison coordinate	inserted / evolved in the native abundance network	Not an endpoint-fit factor or reaction-simplex coordinate
Network-conditioned comparison coordinate	source-predicted and network-pushed-forward	Native network dynamics remain authoritative for the pushforward
Composite comparison tuple	assembled from declared entries	Each entry retains its own registry class and provenance
Frozen comparison input	held fixed	Upstream numerical input, not a native invariant
Fixed comparison setting	imposed / fixed by protocol	Neither derived nor fit as a theory coordinate
Pipeline result	returned / measured by the named pipeline	Numerical evidence, not a theory parameter
Diagnostic / open derivation	diagnostically supported	Independent derivation or full validation remains open

A survey-specific comparison can be written schematically as

$$\Theta_\pi \longrightarrow m_s(\Theta_\pi, \nu_s)\,, \tag{52}$$

where Θ_π is the fixed CPTG branch and ν_s contains declared observational or pipeline quantities; these define the comparison coordinate without becoming CPTG-native parameters [44].

Quantity	Defining relation or value	Class	Role
p_C	π	Native invariant	Curvature exponent
$p_{\rm ac}$	$3-\pi/100 \simeq 2.968584073464$	Derived branch quantity	Acoustic-projection exponent
$\mathcal{B}_\pi$	$1-1/\pi \simeq 0.681690113816$	Native invariant	Late/geometric weight
$\mathcal{T}_\pi$	$1/\pi \simeq 0.318309886184$	Native invariant	Transport weight
$\mathcal{G}_T$	$p_{\rm ac}/p_C \simeq 0.9449296585513721$	Derived branch quantity	Transport factor
$\sqrt{\mathcal{G}_T}$	0.9720749243506758	Derived branch quantity	Hubble-rate transport
$\mathcal{G}_T^2$	0.8928920596100127	Derived branch quantity	Two-step visibility gate
$a_{\pi,0}$	$\pi^{3/2}10^{-10} = 5.568327996831708\times10^{-10}\ \mathrm{m\,s^{-2}}$	Native invariant	Acceleration anchor
N_π	$\ln a$	Derived branch quantity	Native cosmological phase; plain N is reserved for bound-system Structural Mode
$a_\pi(a)$	$a_{\pi,0}E_\pi(a)$	Derived branch quantity	Native acceleration at the selected phase
H_∞	$a_{\pi,0}/c = 57.3131992073\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived branch quantity	Asymptotic Hubble scale
$H_0^{(\pi)}$	$H_\infty/\sqrt{\mathcal{B}_\pi} = 69.4162507897\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived branch quantity	Raw present normalization
R_{C0}	$48\pi c^2/a_{\pi,0} = 788.7823091477$ Gpc	Derived branch quantity	Present native curvature scale
$R_C(a)$	$R_{C0}/E_\pi(a)$	Derived branch quantity	Native curvature-radius evolution
$t_{0,\pi}$	12.754121032276 Gyr	Derived branch quantity	Present native branch time
$\Theta_{\pi T}(a)$	$T/T_0 = 1+z = a^{-1}$, $T_0 = 2.72548$ K	Derived branch quantity	Native thermal property; not a separate branch

Table 5: Native and directly derived geometric quantities. Decimal values are displayed numerical representations of the analytic definitions.

Table 6: Background, clock, and distance-layer quantities.

Quantity	Defining relation or value	Class	Role
$E_\pi(a)^2$	$\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}$	Derived branch quantity	Native expansion
$E_{\rm ac}(a)^2$	$\mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\rm ac}}$	Derived branch quantity	Exact acoustic projection
$H_{0,\rm ac}^{\rm CPTG} \equiv H_{0,\rm CMB}^{\rm CPTG}$	$H_0^{(\pi)}\sqrt{\mathcal{G}_T} = 67.4777967351\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived comparison coordinate	Acoustic/CMB Hubble normalization
$H_{\infty,\rm ac}^{\rm CPTG} \equiv H_{\infty,\rm CMB}^{\rm CPTG}$	$H_\infty\sqrt{\mathcal{G}_T} = 55.7127237837\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived comparison coordinate	Acoustic/CMB asymptote
$\Omega_{m,\rm ac}$	$p_{\rm ac}/(3\pi) \simeq 0.314976552850$	Derived comparison coordinate	Acoustic comparison weight
$\mathcal{R}_3$	$\Omega_{m,\rm ac}/\mathcal{G}_T = 1/3$	Derived comparison coordinate	Exact acoustic rank quotient; $\mathcal{R}_3^2 = 1/9$ supplies the acoustic derivation of the luminosity increment
$\Omega_{\Lambda,\rm ac}$	$1-p_{\rm ac}/(3\pi) \simeq 0.685023447150$	Derived comparison coordinate	Complementary acoustic comparison weight
$t_{0,\rm ac}^{\rm CPTG}$	13.88517358 Gyr, from the acoustic-projection age integral	Derived comparison coordinate	Acoustic branch time coordinate
$E_{{\rm ac},\rho}(a)^2$	$\Omega_{\Lambda,\rm ac} + \Omega_{m,\rm ac}a^{-p_{\rm ac}}$	Diagnostic / open derivation	Density-weighted acoustic overlay; not a branch
$t_{0,{\rm ac},\rho}^{\rm CPTG}$	13.926455176722689 Gyr, from the acoustic-density diagnostic integral	Diagnostic / open derivation	Density-weighted acoustic age diagnostic; not a branch clock
$p_{\rm pol}$	1/3	Parent geometric input	Polarization homogeneity inherited from the parent CPTG framework
$W_\star$	1/3	Derived comparison coordinate	Finite stationary-window value of the declared exterior-safe reduced completion
$\mathfrak{K}_\star$	1/2	Derived comparison coordinate	Dimensionless pure-gradient coordinate at the matched stationary point

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Quantity	Defining relation or value	Class	Role
$\mathsf{C}_{\Pi,\star}$	$I/9$	Frame-conditional derived comparison response	Exact matched isotropic carrier-normalized polarization response within the declared structural frame and exterior-safe reduced completion
$\mathsf{Q}_L^{\mathrm{aff}}$	$I + \mathsf{C}_\Pi$	Registered comparison prescription	Affine luminosity comparison map; not a unique microscopic luminosity operator or universal physical-metric rescaling
$R_{C,L}^{\mathrm{eff,aff}}(a)$	$\sqrt{9/10}\, R_C(a)$	Derived comparison coordinate	Reciprocal effective-radius coordinate used by the affine luminosity rate map
$\mathcal{G}_L^{\mathrm{aff}}$	$10/9$	Derived comparison coordinate	Scalar matched normalization; the same $1/9$ increment is obtained from $\mathsf{C}_{\Pi,\star} = I/9$, $C_H/C_R = 1/9$, and $\mathcal{R}_3^2 = 1/9$ before affine registration
Γ_L^{aff}	$\sqrt{10/9} = 1.054092553389$	Derived comparison coordinate	Matched luminosity rate factor
Ξ_L^{aff}	$\sqrt{9/10} = 0.948683298051$	Derived comparison coordinate	Reciprocal matched rate, radius, and comparison-clock factor
$H_{0,L}^{\mathrm{aff}}$	$H_0^{(\pi)}\sqrt{10/9} = 73.1711530416\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived comparison coordinate	Affine matched luminosity endpoint
$E_L^{\mathrm{aff}}(a)$	$E_\pi(a)$	Derived comparison coordinate	Exact normalized shape of the matched-normalization luminosity trajectory
$t_{0,L}^{\mathrm{aff}}$	12.0996216046 Gyr	Derived comparison coordinate	Present affine luminosity comparison-clock coordinate
$\Gamma_{\mathrm{loc}}^{\mathrm{SH0ES}}$	$1.054075942846 \pm 0.012389029805$	Data-conditioned comparison coordinate	Four-anchor SH0ES luminosity-rate factor; comparison only
$H_{0,\mathrm{lum}}^{\mathrm{anchor}}$	$73.17 \pm 0.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Data-conditioned comparison coordinate	SH0ES anchor used by the historical luminosity scaffold
$\Xi_{\mathrm{lum}}^{\mathrm{anchor}}(a)$	$H_\pi(a)/[H_{0,\mathrm{lum}}^{\mathrm{anchor}} E_{\mathrm{ac},\rho}(a)]$	Data-conditioned comparison coordinate	Phase-dependent SH0ES-conditioned luminosity clock map
$t_{0,\mathrm{lum}}^{\mathrm{anchor}}$	12.8430574232 Gyr	Data-conditioned comparison coordinate	Present luminosity time in the historical conditioned scaffold
ω_{AL}	$1/2$	Registered comparison prescription	Symmetric acoustic–luminosity midpoint weight; not a third branch parameter
$\mathcal{M}_{\mathrm{AL},0}^{\mathrm{aff}}$	$\frac{1}{2}[\sqrt{\mathcal{G}_T} + \sqrt{10/9}] = 1.013083738870$	Derived comparison coordinate	Present theory-constructed acoustic–luminosity midpoint factor
$H_{0,\mathrm{mid}}^{\mathrm{aff}}$	$70.3244748884\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived comparison coordinate	Present affine acoustic–luminosity midpoint rate
$\mathcal{M}_{\mathrm{AL},0}^{\mathrm{anchor}}$	$1.013075433598 \pm 0.006194514903$	Data-conditioned comparison coordinate	Historical midpoint realization using the SH0ES anchor endpoint
$H_{0,\mathrm{mid}}^{\mathrm{anchor}}$	$70.3238983675 \pm 0.43\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Data-conditioned comparison coordinate	Historical midpoint rate using the SH0ES anchor endpoint
$\chi_\star$	13876.8219 Mpc at $z_\star = 1089.92$	Derived comparison coordinate	Acoustic comparison distance; native $H_0^{(\pi)}$ supplies the dimensional prefactor
$L_\star/R_C$	0.53792156 at $z_\star = 1089.92$	Derived comparison coordinate	Native radius-to-radius horizon-containment ratio
z_{enter}	3231.635355	Derived comparison coordinate	Common-phase entry where $L_\star = R_C$

Table 7: Baryon, radiation, primordial-network, nuclear-response, and CMB comparison-map quantities. Frozen inputs are fixed before evaluation but are not relabeled as native geometric invariants. The live mass-seven quantities are abundance-transport coordinates. The first-order nuclear response $\mathcal{C}_{r,\mathcal{N}}^{(1)}$ is network-conditioned because its pushforward is supplied by the declared native reaction dynamics, while SkyPi uses the registered helium-updated source and frozen low-multipole power factors as its current CMB comparison construction. The earlier generic map-envelope operator $\mathcal{E}_\ell^M$ remains documented as a legacy data-conditioned coordinate.

Quantity	Relation or numerical value	Class	Role
f_b	$\pi/20 \simeq 0.157079632679$	Derived comparison coordinate	Baryon fraction
$(\Omega_b h^2)_{\mathrm{ac}}$	0.0225278574944	Frozen comparison input	Acoustic/CMB baryon density
η_{10}^{ac}	6.170380167710	Derived comparison coordinate	Acoustic baryon-to-photon value
$(\Omega_b h^2)_{\mathrm{BBN}}$	0.021898765370	Derived comparison coordinate	Transported BBN baryon density

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Quantity	Relation or numerical value	Class	Role
η_{10}^{BBN}	$\eta_{10}^{\mathrm{ac}}\sqrt{\mathcal{G}_T} = 5.998071834744$	Derived comparison coordinate	BBN baryon-to-photon value
$N_{\mathrm{eff}}^{\mathrm{CPTG}}$	$p_{\mathrm{ac}} \simeq 2.968584073464$	Derived comparison coordinate	Radiation coordinate
$\Theta_{\mathrm{BBN}}^{\mathrm{CPTG}}$	$\left(\eta_{10}^{\mathrm{BBN}}, N_{\mathrm{eff}}^{\mathrm{CPTG}}, \tau_7\right)$	Composite comparison tuple	Primordial-network input tuple; entries retain their individual classes
τ_7	$\ln \pi \simeq 1.1447298858494002$	Abundance-transport comparison coordinate	Live mass-seven optical depth
$\mathcal{S}_7$	$e^{-\tau_7} = 1/\pi \simeq 0.318309886184$	Abundance-transport comparison coordinate	Isolated mass-seven survival
$\mathcal{C}_{r,\mathcal{N}}^{(1)}$	$\mathsf{P}_3\mathsf{L}_{\mathcal{N}}^{(1)}(\eta_{10})\mathcal{A}_{r,\mathcal{N}}^{(1)}\mathbf{s}_r$	Network-conditioned comparison coordinate	First-order shared nuclear endpoint response
$\mathcal{D}_c$	0.9735353159758136	Derived comparison coordinate	Cold-sector CMB transport
$\Omega_{c,\mathrm{CPTG}}h^2$	0.120885025627	Frozen comparison input	Pre-map cold-sector input
$\Omega_{c,\mathrm{eff}}h^2$	0.117685841620526	Derived comparison coordinate	CMB cold density
$A_{s,\mathrm{CPTG}}$	2.04×10^{-9}	Frozen comparison input	Pre-map scalar amplitude
$\mathcal{A}_s$	$\pi/3$	Derived comparison coordinate	Scalar-amplitude gate
$A_{s,\mathrm{eff}}$	$2.136283004441 \times 10^{-9}$	Derived comparison coordinate	CMB scalar amplitude
$\tau_{\mathrm{reio,CPTG}}$	0.066	Frozen comparison input	Pre-map reionization optical depth
$\mathcal{T}_\tau$	$\mathcal{G}_T^2$	Derived comparison coordinate	Visibility gate
$\tau_{\mathrm{reio,eff}}$	0.05893087593426084	Derived comparison coordinate	CMB reionization optical depth
n_s	$1 - \pi/100 \simeq 0.968584073464$	Derived comparison coordinate	Scalar tilt
$Y_{\mathrm{He,CMB}}$	0.24576024693340703	Frozen comparison input	Current SkyPi CMB helium source; gate-off PRIMAT Y_p
A_{lens}	1	Fixed comparison setting	No phenomenological lensing inflation
q_ℓ^2	Piecewise fixed low-ℓ power factors; Eq. (97)	Frozen, observation-conditioned comparison input	SkyPi source-power mapping before beam/mask forward evaluation
$\mathcal{E}_\ell^M$	$\left(C_\ell^M/C_\ell^{\mathrm{obs}}\right)^{1/2}$	Legacy data-conditioned comparison coordinate	Historical phase-locked map-envelope transport

Quantity	Defining relation or value	Class	Role
$S_{12}^{\mathrm{CPTG}}(0)$	$K_{S0}I_{C7} = 0.2145003969219307$ eV b	Derived comparison coordinate	Zero-energy reaction strength
$S_{12}^{\mathrm{CPTG}}(E)$	$S_{12}^{\mathrm{CPTG}}(0)F_{\mathrm{ext}}(E)$	Data-conditioned comparison coordinate	Declared energy-shape projection
$\mathcal{R}_{pD}^{\mathrm{CPTG}}(T)$	Maxwellian average of $S_{12}^{\mathrm{CPTG}}(E)$	Data-conditioned comparison coordinate	Thermonuclear-rate projection

Table 8: Commissioning-channel reaction-observable coordinates. The zero-energy normalization is fixed by the CPTG commissioning construction. The finite-energy $S(E)$ and thermal-rate coordinates retain the declared external energy-shape carrier and are therefore data-conditioned rather than CPTG-native energy-shape invariants.

Quantity	Relation or value	Class	Role
A_L^{CPTG}	$(p_C/p_{\mathrm{ac}})^2 \simeq 1.119956202138$	Derived branch quantity	Branch-derived lensing response scale
$C_T(a,k)$	$\Phi_\pi(a,k)/\Phi_0(k)$	Bridge-validated comparison coordinate	Curvature-transport amplitude response
$\mathcal{R}_T(a,k)$	$P_\pi(a,k)/P_0(a,k) = C_T(a,k)^2$	Bridge-validated comparison coordinate	Locked power-spectrum response map
$q_{\mathrm{bandpower}}$	$(1-\mathcal{G}_T)/\mathcal{B}_\pi$	Registered comparison prescription	CMB-lensing bandpower projection weight
$A_{\mathrm{bandpower}}$	1.009690662776	Derived comparison coordinate	Raw CMB-lensing projection
$S_{8,\mathrm{anchor}}$	0.834	Frozen comparison input	Planck-derived amplitude normalization used by the locked late-lensing transport
$S_{8,\mathrm{late}}^{\mathrm{CPTG}}$	$\mathcal{G}_T S_{8,\mathrm{anchor}} = 0.788071335232$	Transported comparison coordinate	Late galaxy-lensing comparison
α_C	$1-2/\pi$	Registered comparison prescription	CMB-lensing compression exponent
$S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}}$	0.821191065949	Derived comparison coordinate	CMB-lensing compression
$q_\perp, q_\parallel$	AP distance/ruler ratios	Derived comparison coordinates	DESI/BAO
$q_{\mathrm{iso}}, q_{\mathrm{ap}}$	$(q_\perp^2 q_\parallel)^{1/3}$, $q_\parallel/q_\perp$	Derived comparison coordinates	ShapeFit
d_f	$f_{\mathrm{CPTG}}/f_{\mathrm{fid}}$	Derived comparison coordinate	ShapeFit growth
d_m	0 in the declared adapter	Fixed comparison setting	ShapeFit tilt placeholder
r_d^{ratio}	$\mathcal{G}_T^{-1/4} \simeq 1.014261942163$	Diagnostic / open derivation	Quarter-ruler candidate
r_d	148.998491 Mpc	Frozen comparison input	Externally fitted comparison-layer BAO normalization

Table 9: Growth, lensing, curvature-transport spectrum, ShapeFit, and ruler coordinates. The Route B pair C_T and $\mathcal{R}_T = C_T^2$ is bridge-validated at the comparison-coordinate level; this status does not claim a completed native spectral-response implementation. The late galaxy-lensing amplitude is a transported comparison coordinate: the frozen amplitude normalization is fixed before comparison and the CPTG map is carried by $\mathcal{G}_T$. The quarter-ruler relation remains a supported diagnostic candidate rather than a completed independent derivation.

The numerical inputs and fixed protocol settings used by downstream comparisons have the following upstream provenance. Where the present article carries a value from the locked comparison construction rather than deriving it analytically, that limitation is stated explicitly.

Quantity / setting	Upstream provenance	Freeze rule	Claim boundary / overlap
$(\Omega_b h^2)_{\mathrm{ac}} = 0.0225278574944$	Locked CPTG acoustic/CMB comparison-row construction and reproducibility materials [51]	Fixed before CMB and primordial-network evaluations; no sector-specific retuning	Frozen comparison input; not promoted to an analytic native invariant
$\Omega_{c,\mathrm{CPTG}} h^2 = 0.120885025627$	Locked pre-map CPTG CMB comparison row [51]	Fixed before every cited CMB likelihood evaluation	Frozen comparison input; CPTG subscript denotes row assignment, not native derivation
$A_{s,\mathrm{CPTG}} = 2.04 \times 10^{-9}$	Locked pre-map CPTG CMB comparison row [51]	Fixed before every cited CMB likelihood evaluation	Frozen comparison input; the derived quantity is $A_{s,\mathrm{eff}}$
$\tau_{\mathrm{reio,CPTG}} = 0.066$	Locked pre-map CPTG CMB comparison row [51]	Fixed before low-ℓ, high-ℓ, and combined-stack tests	Frozen comparison input; the derived quantity is $\tau_{\mathrm{reio,eff}}$
$Y_{\mathrm{He,CMB}} = 0.24576024693340703$	Frozen gate-off CPTG PRIMAT abundance [43]	Fixed for current SkyPi source; no per-map adjustment	Frozen CMB comparison input, not a native helium invariant; historical registry used 0.2485
$A_{\mathrm{lens}} = 1$	Physical-consistency setting of the CMB comparison protocol [3]	Imposed in every cited CMB evaluation	Fixed comparison setting; distinct from branch-derived A_L^{CPTG}
$S_{8,\mathrm{anchor}} = 0.834$	Planck-derived amplitude normalization [2]	Frozen before DES, HSC, and KiDS late-lensing comparisons	Transport anchor; CMB-data overlap is declared where a CMB-based downstream comparison is used
$d_m = 0$	Declared ShapeFit adapter choice	Fixed throughout the compressed adapter	Fixed comparison setting; no CPTG spectral-tilt prediction is claimed
$r_d = 148.998491\,\mathrm{Mpc}$	Externally fitted comparison-layer normalization of the existing compressed BAO implementation	Fixed for that compressed BAO calculation; not reclassified by the quarter-ruler candidate	Frozen comparison input; no native absolute-ruler derivation is claimed

Table 10: Provenance and freeze rules for the paper's frozen comparison inputs and fixed comparison settings.

The symbols $\Omega_{c,\mathrm{CPTG}} h^2$, $A_{s,\mathrm{CPTG}}$, and $\tau_{\mathrm{reio,CPTG}}$ retain the CPTG subscript because they belong to the CPTG pre-map

comparison row. The subscript does not change their registry class: they remain frozen comparison inputs, whereas $\Omega_{c,\mathrm{eff}}h^2$, $A_{s,\mathrm{eff}}$, and $\tau_{\mathrm{reio,eff}}$ are the derived comparison coordinates produced by the stated gates.

Likelihood values, χ^2 values, posterior diagnostics, mask row counts, and scan optima are *pipeline results*; they are intentionally excluded from the locked-parameter registry and are reported only in the validation sections together with their method and provenance.

5.1 No-Retuning Rule for Comparison Coordinates

A comparison-layer calculation should follow four steps:

$$\begin{gathered}\text{fix the CPTG-native branch}\\ \downarrow\\ \text{choose the observational coordinate system}\\ \downarrow\\ \text{apply the locked transport map}\\ \downarrow\\ \text{compare without sector-by-sector retuning.}\end{gathered} \tag{53}$$

This rule prevents the comparison layer from becoming a hidden fitting layer. If a parameter is allowed to move during a likelihood calculation, it must be identified as a nuisance or control parameter of the observational pipeline, not as a new CPTG branch parameter.

6 Microscopic Nuclear and Primordial BBN Comparison Layer

The CPTG baryon fraction is fixed by

$$f_b = \frac{\pi}{20} \simeq 0.157079632679. \tag{54}$$

The acoustic/CMB physical baryon-density comparison input is

$$(\Omega_b h^2)_{\mathrm{ac}} = 0.0225278574944. \tag{55}$$

The analysis uses the standard fixed-temperature conversion convention [13]

$$\eta_{10} \simeq 273.9\,\Omega_b h^2, \tag{56}$$

so that the stored-precision acoustic baryon-density coordinate gives

$$\eta_{10}^{\mathrm{ac}} = 6.170380167710. \tag{57}$$

The displayed $(\Omega_b h^2)_{\mathrm{ac}}$ value is rounded relative to the stored numerical value used by the conversion, so reconstruction from the printed decimal can differ only in the last displayed digits.

The effective-radiation comparison coordinate is identified with the acoustic exponent,

$$N_{\mathrm{eff}}^{\mathrm{CPTG}} = p_{\mathrm{ac}} \simeq 2.968584073464. \tag{58}$$

For light-element comparison, the baryon coordinate transports as

$$\eta_{10}^{\mathrm{BBN}} = \eta_{10}^{\mathrm{ac}}\sqrt{\mathcal{G}_T} = 5.998071834744, \tag{59}$$

$$(\Omega_b h^2)_{\mathrm{BBN}} = (\Omega_b h^2)_{\mathrm{ac}}\sqrt{\mathcal{G}_T} = 0.021898765370. \tag{60}$$

Thus

$$((\Omega_b h^2)_{\mathrm{ac}}, \eta_{10}^{\mathrm{ac}}) \longrightarrow ((\Omega_b h^2)_{\mathrm{BBN}}, \eta_{10}^{\mathrm{BBN}}) \tag{61}$$

is a comparison-coordinate transport rather than a refit of primordial abundances.

For reference, the Standard Model radiation benchmark used by the modern BBN literature is $N_{\mathrm{eff}}^{\mathrm{SM}} = 3.044$ [25], while Planck combined with BAO constrains the corresponding observational extension to $N_{\mathrm{eff}} = 2.99 \pm 0.17$ [2]. These are different objects: the former is a theoretical Standard Model benchmark and the latter is an observational constraint. They are therefore not combined into a hybrid "central value plus error."

6.1 Commissioning-Channel Reaction-Observable Coordinates

The commissioned $D(p,\gamma)^3\mathrm{He}$ calculation supplies a direct bridge from a reaction-specific CPTG source construction into conventional nuclear-astrophysics observables. Its ordering is

$$\begin{gathered}\text{reaction source amplitude}\\ \downarrow\\ \text{dimensionless source intensity}\\ \downarrow\\ S_{12}^{\mathrm{CPTG}}(0)\\ \downarrow\\ S_{12}^{\mathrm{CPTG}}(E)\\ \downarrow\\ \mathcal{R}_{pD}^{\mathrm{CPTG}}(T).\end{gathered} \tag{62}$$

The retained coherent commissioning amplitude is $\mathcal{A}_{C6} = 17.5595746398194$, giving the dimensionless native source intensity

$$I_{C7} = |\mathcal{A}_{C6}|^2 = 308.3386615313886. \tag{63}$$

The amplitude contraction itself is reaction-specific and belongs to the commissioning construction; the comparison-coordinate step begins when this native dimensionless intensity is converted into an observable with units [15].

For the zero-energy reaction-strength coordinate, the dimensional bridge is

$$K_{S0} = \frac{\alpha\pi}{33 - 1/(7\pi)}\ \mathrm{eV\,b} = 0.0006956649414530029\ \mathrm{eV\,b}, \tag{64}$$

so that

$$S_{12}^{\mathrm{CPTG}}(0) \equiv K_{S0} I_{C7} = 0.2145003969219307\ \mathrm{eV\,b}. \tag{65}$$

The support-count denominator in K_{S0} is fixed by the commissioned $A = 3$ construction before external S-factor comparison; it is not fitted to the LUNA extrapolation. The corresponding observational checks are recorded later in Section 12.1.

The finite-energy coordinate requires a stricter classification. The commissioned construction fixes the zero-energy normalization, but its low-energy shape is supplied by a declared external carrier. Write

$$S_{12}^{\mathrm{CPTG}}(E) = S_{12}^{\mathrm{CPTG}}(0) F_{\mathrm{ext}}(E). \tag{66}$$

For the Solar Fusion II low-energy carrier used in the commissioning analysis,

$$F_{\mathrm{SFII}}(E) = \frac{S_{12,\mathrm{SFII}}(E)}{S_{12,\mathrm{SFII}}(0)}, \tag{67}$$

with

$$S_{12,\mathrm{SFII}}(E) = \left[0.2145 + 5.5612\left(\frac{E}{\mathrm{MeV}}\right) + 4.6581\left(\frac{E}{\mathrm{MeV}}\right)^2\right] \mathrm{eV\,b}. \tag{68}$$

At $E_0 = 6.64\,\mathrm{keV}$, $F_{\mathrm{SFII}} = 1.1731083532203264$ and

$$S_{12}^{\mathrm{CPTG}}(E_0) = 0.2516322073981925\ \mathrm{eV\,b}, \tag{69}$$

The energy-shape function is part of the declared comparison coordinate and is not relabeled as a new CPTG-native invariant. Its observational comparison is recorded in Section 12.1.

For charged-particle capture,

$$\sigma(E) = \frac{S(E)}{E}\exp[-2\pi\eta(E)], \qquad \eta(E) = Z_1 Z_2 \alpha\sqrt{\frac{\mu c^2}{2E}}, \tag{70}$$

with $Z_1 = Z_2 = 1$ and $\mu = m_p m_D/(m_p + m_D)$ for this channel. The thermonuclear comparison coordinate is the Maxwellian pushforward

$$\mathcal{R}_{pD}^{\mathrm{CPTG}}(T) \equiv N_A\left(\frac{8}{\pi\mu}\right)^{1/2}\frac{1}{(k_B T)^{3/2}}\int_0^\infty S_{12}^{\mathrm{CPTG}}(E)\exp\left[-\frac{E}{k_B T} - 2\pi\eta(E)\right] dE. \tag{71}$$

Because this thermal coordinate inherits the declared $S(E)$ shape carrier, it is data-conditioned at finite energy even though its CPTG zero-energy normalization is fixed. A tabulated network may implement the same normalization through

$$R_{pD}^{\mathrm{CPTG}}(T) = \lambda_{pD} R_{pD}^{\mathrm{table}}(T), \qquad \lambda_{pD} = 1.0000018504518913, \tag{72}$$

where $\lambda_{pD} = S_{12}^{\mathrm{CPTG}}(0)/S_{12}^{\mathrm{table}}(0)$ with $S_{12}^{\mathrm{table}}(0) = 0.2145000000000000\,\mathrm{eV\,b}$. This ratio is an implementation adapter, not a new geometric parameter [15].

The completed commissioning-channel observational checks are collected in Section 12.1; they do not modify the construction above.

6.2 Network-Conditioned Nuclear Reaction-Response Pushforward

The primordial input tuple above specifies the background and the separate live mass-seven transport. The deeper reaction-response comparison begins from the universal four-sector source geometry rather than from an endpoint abundance vector.

Let the ordered four-sector baryon coordinate be

$$\mathbf{X} = \begin{pmatrix} X_V \\ X_B \\ X_C \\ X_S \end{pmatrix}, \tag{73}$$

for vertex, bridge, closure, and saturation. During the active light-element reaction epoch these grouped baryon coordinates are

$$X_V = Y_n + Y_p^{\mathrm{free}}, \qquad X_B = 2Y_D, \qquad X_C = 3\left(Y_T + Y_{^3\mathrm{He}}\right), \qquad X_S = 4Y_{^4\mathrm{He}}. \tag{74}$$

For a declared reaction-rate perturbation, define the logarithmic rate coordinate before applying the incidence operator,

$$x \equiv -\ln f_r. \tag{75}$$

The three directed transport edges are collected in

$$\mathbf{j} = \begin{pmatrix} j_{VB} \\ j_{BC} \\ j_{CS} \end{pmatrix}, \tag{76}$$

and the ordered topology fixes the exact incidence operator

$$\mathsf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \frac{d\mathbf{X}}{dx} = \mathsf{D}\,\mathbf{j}. \tag{77}$$

The matrix D is fixed by the vertex-to-bridge-to-closure-to-saturation topology; it is not a fitted network matrix. Its columns automatically lie in the baryon-conserving tangent subspace because each column sums to zero [22].

The coarse four-sector coordinate $\mathbf{X}$ records baryon transport between vertex, bridge, closure, and saturation, but the active reaction state also retains the internal charge orientations of the vertex and closure sectors. Define

$$P_V = Y_p^{\mathrm{free}} - Y_n, \qquad P_C = 3\left(Y_{^3\mathrm{He}} - Y_T\right), \tag{78}$$

so that the resolved source state may be represented by the six-coordinate vector

$$\widetilde{\mathbf{X}} = \begin{pmatrix} X_V \\ P_V \\ X_B \\ X_C \\ P_C \\ X_S \end{pmatrix}. \tag{79}$$

The mapping from the six resolved light-species abundances into these grouped transport and charge-orientation coordinates is invertible within the declared four-sector backbone [22]. Here Y_p^{free} denotes the free-proton species coordinate during network evolution. It is not interchangeable with the final reported D/H number ratio, which is a separate network endpoint observable. Baryon conservation and charge conservation remove two tangent directions from this six-coordinate state, leaving four independent source-space directions. Three are the ordered transport edges carried by D; the fourth is the charge-constrained polarization direction

$$3\,\delta P_V + \delta P_C = 0, \qquad \Pi = \delta P_V = -\frac{\delta P_C}{3}. \tag{80}$$

so that the exact reaction-source coordinate is

$$\mathbf{s}_r = \begin{pmatrix} q_{VB} \\ q_{BC} \\ q_{CS} \\ \Pi \end{pmatrix}_r . \tag{81}$$

Here $\mathbf{j} = (j_{VB}, j_{BC}, j_{CS})^T$ is the generic edge-current coordinate on which the incidence operator acts, whereas (q_{VB}, q_{BC}, q_{CS}) are the reaction-source coordinates of reaction r along those same ordered edge directions. The distinction separates the topology of the universal incidence operator from the reaction-specific source direction.

Reaction stoichiometry fixes this source direction before numerical endpoint responses are opened; it is not a fitted abundance direction [22]. The three incidence-edge coordinates together with the single independent polarization coordinate close the declared reaction-source tangent space at exact rank four. This rank statement belongs to the source geometry; it does not assert that every network endpoint response is four-dimensional or that different native networks share one numerical endpoint matrix.

A native network $\mathcal{N}$ supplies its own code-local source kernel. At first order in the logarithmic rate coordinate x defined in Equation (75),

$$\mathcal{F}_{r,\mathcal{N}}(x) = \mathcal{A}^{(1)}_{r,\mathcal{N}} x + \mathcal{O}(x^2), \tag{82}$$

which generates the source drive

$$\mathbf{d}^{(1)}_{r,\mathcal{N}} = \mathcal{A}^{(1)}_{r,\mathcal{N}} \mathbf{s}_r. \tag{83}$$

The native network dynamics then define the observable pushforward

$$\mathbf{y}^{(1)}_{r,\mathcal{N}} = \mathsf{L}^{(1)}_{\mathcal{N}}(\eta_{10}) \mathbf{d}^{(1)}_{r,\mathcal{N}}. \tag{84}$$

Neither the numerical entries nor even the declared observable dimension of $\mathsf{L}^{(1)}_{\mathcal{N}}$ are assumed to be identical between PRIMAT, PArthENoPE, and PRyMordial. The universal object is the fixed source geometry and factorization; the numerical pushforward remains network-local [22].

For the shared primordial endpoint comparison used in the present authority chain, define

$$\mathsf{P}_3 \mathbf{y} = \begin{pmatrix} \Delta \ln(D/H) \\ \Delta \ln({}^3\mathrm{He}/H) \\ \Delta \ln({}^7\mathrm{Li}/H) \end{pmatrix}. \tag{85}$$

The first-order network-conditioned nuclear comparison coordinate is therefore

$$\mathcal{C}^{(1)}_{r,\mathcal{N}}(\eta_{10}) \equiv \mathsf{P}_3 \mathbf{y}^{(1)}_{r,\mathcal{N}} = \mathsf{P}_3 \mathsf{L}^{(1)}_{\mathcal{N}}(\eta_{10}) \mathcal{A}^{(1)}_{r,\mathcal{N}} \mathbf{s}_r. \tag{86}$$

This is a comparison coordinate, not a new native invariant. It depends on the fixed CPTG source direction, the declared native network, the baryon-density coordinate, and the chosen observable projection. The bare source vector $\mathbf{s}_r$ is therefore not identified directly with the final abundance-response vector. The completed PRIMAT source/Jacobian campaign validates the dynamical pushforward itself, rather than a direct source-to-endpoint identification [22, 14].

6.3 Primordial Nuclear and Mass-Seven Abundance-Transport Coordinates

The comparison layer supplied to a native primordial network is conveniently collected as

$$\Theta^{\mathrm{CPTG}}_{\mathrm{BBN}} = \left(\eta^{\mathrm{BBN}}_{10}, N^{\mathrm{CPTG}}_{\mathrm{eff}}, \tau_7\right). \tag{87}$$

The first two entries are the transported baryon and radiation coordinates above. The third is a distinct live abundance-transport coordinate. In the geometric pi-branch comparison construction, the integrated mass-seven transport depth is fixed as

$$\tau_7 \equiv \int \Gamma_{7,\mathrm{trans}}(t)\, dt = \ln \pi \simeq 1.1447298858494002. \tag{88}$$

The corresponding isolated-channel survival coordinate is therefore

$$\mathcal{S}_7 \equiv e^{-\tau_7} = \frac{1}{\pi} \simeq 0.3183098861837907. \tag{89}$$

The pi-branch comparison construction fixes $\tau_7 = \ln \pi$ before the downstream nuclear-reaction and lithium validations. The downstream nuclear-reaction and cosmological-lithium analyses apply and test that already-fixed law rather than deriving it from the lithium endpoint [15, 22, 23].

The live network action is not a post-processing division of a completed abundance. For the two surviving mass-seven channels it has the comparison-level form

$$\frac{dY_i}{dt} \longrightarrow \left(\frac{dY_i}{dt}\right)_{\mathrm{BBN}} - \Gamma_{7,\mathrm{trans}}(t) Y_i, \qquad i \in \left\{{}^7\mathrm{Li}, {}^7\mathrm{Be}\right\}. \tag{90}$$

The exact relation $\mathcal{S}_7 = 1/\pi$ is the isolated survival law. A live native network continues ordinary production, destruction, and redistribution while the transport term is active, so the measured gate-off/gate-on endpoint ratio is a network-response quantity and is tested against, rather than algebraically forced to equal, $1/\pi$.

The four-sector CPTG nuclear source geometry remains a separate parent structure: free nucleons form the vertex, deuterium the bridge, tritium/helium-3 the closure sector, and helium-4 the saturation sector. Mass seven is not promoted to a fifth baryon-simplex coordinate. The $A = 7$ law is a distinct downstream abundance-transport action on the evolving ^{7}Li and ^{7}Be variables, not an asserted conventional nuclear-reaction channel [22].

7 Fixed CMB Acoustic Transport Layer

The current CMB source-construction authority in this reference is CPTG SkyPi [42]. The earlier Planck/WMAP comparison-map implementation is retained only as a legacy test, not as a competing current prescription. The native Pi relations, acoustic projection, and other registered comparison coordinates remain unchanged. The CPTG acoustic/CMB projection is fixed before likelihood evaluation. The native quantity p_C, the derived acoustic quantities p_{ac}, $\mathcal{G}_T$, and $\sqrt{\mathcal{G}_T}$, together with the CMB inputs and output coordinates, are registered in Tables 5, 7, and 10. The active native-to-comparison map is

$$\Omega_{c,\mathrm{eff}} h^2 = \Omega_{c,\mathrm{CPTG}} h^2\, \mathcal{D}_c, \tag{91}$$

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}}\, \mathcal{A}_s, \tag{92}$$

$$\tau_{\mathrm{reio,eff}} = \tau_{\mathrm{reio,CPTG}}\, \mathcal{T}_\tau. \tag{93}$$

The cold-sector transport is

$$\mathcal{D}_c = \sqrt{\mathcal{G}_T}\left(1 + \frac{\pi - 3}{30\pi}\right) = 0.9735353159758136, \tag{94}$$

while the amplitude and visibility gates are

$$\mathcal{A}_s = \frac{\pi}{3} = 1.0471975511965976, \qquad \mathcal{T}_\tau = \mathcal{G}_T^2 = 0.8928920596100127. \tag{95}$$

The amplitude and visibility gates are derived explicitly from the registered branch quantities in Section 7.2. The cold-sector factor $\mathcal{D}_c$ retains its registry status as the derived cold-sector transport coordinate. The arithmetic closure of all three mapped quantities against the fixed row is tested in Section 13.10.

Parameter	Fixed comparison value	Registry class
H_0	$67.4777967351\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived comparison coordinate
$\Omega_b h^2$	0.022527857494	Frozen comparison input
$\Omega_c h^2$	0.117685841620526	Derived comparison coordinate
n_s	0.968584073464	Derived comparison coordinate
N_{eff}	2.968584073464	Derived comparison coordinate
$Y_{\mathrm{He,CMB}}$	0.24576024693340703	Frozen SkyPi comparison input
A_s	$2.136283004441 \times 10^{-9}$	Derived comparison coordinate
τ_{reio}	0.058930875934	Derived comparison coordinate
A_{lens}	1	Fixed comparison setting

Table 11: Current CPTG SkyPi CMB acoustic-transport comparison row. The previously published CMB helium registry entry $Y_{\mathrm{He,CMB}} = 0.2485$ belongs to the legacy comparison specification; the original map-closure scripts did not explicitly pass that historical value as a helium override. Earlier likelihood-result records retain their original declared setups.

For arithmetic-precision bookkeeping, the corresponding unrounded mapped values are $\Omega_{c,\mathrm{eff}}h^2 = 0.11768584162052577$, $A_{s,\mathrm{eff}} = 2.1362830044410592 \times 10^{-9}$, and $\tau_{\mathrm{reio,eff}} = 0.05893087593426084$. The table retains the declared comparison-row display precision.

The table above is the current SkyPi comparison-input row, not the original fixed-row likelihood-test configuration. Both share the registered geometric coordinates, but the historical likelihood results retain their own numerical-input and nuisance conventions; they are not retroactively identified as helium-updated SkyPi map tests. The likelihood setting $A_{\mathrm{lens}} = 1$ is distinct from the branch-derived response scale A_L^{CPTG}; no phenomenological $A_{\mathrm{lens}} > 1$ enhancement is introduced [3, 2].

7.1 Authoritative SkyPi CMB Source and Frozen Low-Multipole Factors

The current SkyPi temperature source uses the preceding frozen Pi-branch CMB coordinates with the gate-off PRIMAT helium mass fraction

$$Y_{\mathrm{He,CMB}}^{\mathrm{SkyPi}} = Y_p^{\mathrm{CPTG}} = 0.24576024693340703. \tag{96}$$

This is an explicit, previously frozen CMB-input substitution; the observational helium values are not used to select it. The historical Pi Version II entry 0.2485 is superseded for current SkyPi source calculations. The earlier map-closure scripts did not explicitly override the solver's default helium handling, so the old registry value must not be retroactively attributed to those runs [43, 42, 44].

Let $C_\ell^{\pi,\mathrm{src}}$ denote the raw temperature C_ℓ obtained from the frozen Pi/CMB-facing numerical comparison row with this helium input. SkyPi's current registered source-power construction is

$$C_\ell^{\mathrm{SkyPi,src}} = q_\ell^2\, C_\ell^{\pi,\mathrm{src}}, \qquad q_\ell^2 = \begin{cases} 0.1311364911405572, & \ell = 2, \\ 0.7738684053212608, & 3 \le \ell \le 9, \\ 0.8499508217809941, & 10 \le \ell \le 29, \\ 1, & \ell \ge 30. \end{cases} \tag{97}$$

The q_ℓ^2 values are *power multipliers already squared*; they must not be squared again. Their coefficients were registered using the 31 October 2026 Commander HM1×HM2 observational comparison and thereafter held fixed across runs. They are therefore frozen *observation-conditioned comparison coordinates*, not additional native Pi-geometric identities and not values re-estimated from the selected Planck map. Source multipoles $\ell \ge 30$ are unchanged by this factor; after beam and mask mode coupling, the corresponding *measured* pseudo-spectrum need not be unchanged at those multipoles [42].

The beam, pixel sampling, chosen intensity/OE mask, harmonic window, and optional split processing constitute separately declared Planck measurement operators on this fixed source. SkyPi evaluates the observed temperature spectrum from Planck data independently of the CPTG source, applies the registered response and mask-forward construction, and makes no per-run global amplitude or offset fit. This is a numerical comparison coordinate, not a separate native CPTG perturbation theory. Only the operator definition, not any new map residual table, is specified in this technical reference [42].

7.2 Geometric Origin of the CMB Amplitude and Visibility Gates

The locked CMB comparison layer contains two geometric projection gates beyond the cold-sector distance map. They are derived from the registered branch quantities rather than fitted to CMB residuals.

7.2.1 Amplitude Projection and the $\pi/3$ Curvature Gate

The scalar amplitude comparison is a three-spatial-direction projection of the registered curvature branch $p_C = \pi$. The branch-to-acoustic projection is therefore

$$\mathcal{A}_C = \frac{p_C}{3} = \frac{\pi}{3} = 1.0471975511965976. \tag{98}$$

The comparison map uses $\mathcal{A}_s \equiv \mathcal{A}_C$, so

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}} \frac{\pi}{3}. \tag{99}$$

The frozen value $A_{s,\mathrm{CPTG}} = 2.04 \times 10^{-9}$ is an input to this map, not an independently derived native amplitude; its provenance and freeze rule are given in Table 10.

7.2.2 Visibility Transport and the $\mathcal{G}_T^2$ Reionization-Depth Gate

The reionization-depth comparison coordinate is carried by the second-order acoustic transport visibility projection

$$\mathcal{T}_\tau = \mathcal{G}_T^2 = 0.8928920596100127. \tag{100}$$

The resulting map is

$$\tau_{\mathrm{reio,eff}} = \tau_{\mathrm{reio,CPTG}} \mathcal{G}_T^2. \tag{101}$$

The frozen value $\tau_{\mathrm{reio,CPTG}} = 0.066$ is a pre-map comparison input and is not retuned against low-ℓ polarization. The independent numerical bookkeeping test of both gates, including the effective values $A_{s,\mathrm{eff}} = 2.136283004441059 \times 10^{-9}$ and $\tau_{\mathrm{reio,eff}} = 0.05893087593426084$, is reported in Section 13.10.

8 CMB Comparison Coordinates and Reference Quantities

The locked CPTG CMB row is a fixed geometric construction expressed in the coordinates used by CMB likelihoods. External reference quantities do not all have the same provenance: some are Planck posterior values, some are theoretical Standard Model benchmarks, and some are BBN-consistency-derived or convention-converted quantities. Keeping those classes separate prevents a theoretical benchmark from being misread as a direct posterior measurement.

Quantity	CPTG comparison value	External reference	Reference class
H_0	67.4777967351	67.36	Planck base-ΛCDM posterior [2]
$\Omega_b h^2$	0.022527857494	0.02237	Planck base-ΛCDM posterior [2]
$\Omega_c h^2$	0.117685841621	0.1200	Planck base-ΛCDM posterior [2]
n_s	0.968584073464	0.9649	Planck base-ΛCDM posterior [2]
$N_{\rm eff}$	2.968584073464	3.044	Standard Model BBN benchmark [25]
$N_{\rm eff}$	2.968584073464	2.99 ± 0.17	Planck+BAO one-parameter extension [2]
$Y_{\rm He,CMB}$	0.24576024693340703	$\simeq 0.2454$	Frozen PRIMAT input [43]; Planck BBN-consistency-derived reference [2]
A_s	$2.136283004441 \times 10^{-9}$	2.100549×10^{-9}	Converted Planck scalar-amplitude convention [2]
$\tau_{\rm reio}$	0.058930875934	0.0544	Planck reionization-optical-depth posterior [2]

Table 12: Current SkyPi CMB comparison coordinates and external references, separated by provenance class. Historical likelihood tests are not re-evaluated by this table.

The $N_{\rm eff} = 3.044$ row is a theoretical Standard Model benchmark, while 2.99 ± 0.17 is an observational Planck+BAO constraint in an extended parameter space. They are not combined into a single "measured" reference value. Likewise, the Planck $Y_{\rm He}$ entry is BBN-consistency-derived rather than an independent direct CMB helium measurement.

The role of the external row is to provide reference scale and provenance. The actual fixed-row decision is made from like-for-like likelihood shifts,

$$\Delta\chi^2 = \chi^2_{\rm CPTG,locked} - \chi^2_{\rm control}, \tag{102}$$

under identical likelihood and nuisance machinery. The CPTG coordinates therefore need not numerically equal each external central value one by one in order for the fixed row to remain at control level.

9 Recombination-Scale Spectral Response and CMB Projection Construction

CPTG treats growth and lensing as fixed geometric response channels of the nonlinear curvature-polarization and curvature-transport field. To interface with conventional CMB and lensing likelihoods, this section uses the scalar metric, effective Poisson relation, and Weyl-potential combination below solely as external comparison-coordinate definitions. They are not asserted as CPTG-native field equations, do not make GR or FLRW the governing CPTG framework, and do not introduce a dark-matter sector. The functions $\mu_{\rm CPTG}$ and $\Sigma_{\rm CPTG}$ are translation objects at this boundary. In this comparison coordinate, the scalar metric is written as

$$ds^2 = -(1+2\Psi)\,c^2dt^2 + a(t)^2\,(1-2\Phi)\,d\mathbf{x}^2. \tag{103}$$

The growth response is defined by the effective Poisson relation

$$k^2\Psi = -4\pi G a^2 \bar{\rho}\,\mu_{\rm CPTG}(a,k)\delta. \tag{104}$$

The lensing response is defined through the Weyl-potential combination

$$k^2\,(\Phi+\Psi) = -8\pi G a^2 \bar{\rho}\,\Sigma_{\rm CPTG}(a,k)\delta. \tag{105}$$

9.1 Curvature-Transport Spectrum Bridge

The locked Route B Option 1 comparison coordinate supplies an amplitude-level curvature-transport bridge between the CPTG potential response and a conventional spectrum coordinate. The amplitude comparison coordinate is

$$C_T(a,k) \equiv \frac{\Phi_\pi(a,k)}{\Phi_0(k)}, \tag{106}$$

so that

$$\Phi_\pi(a,k) = \Phi_0(k)\, C_T(a,k). \tag{107}$$

Because a power spectrum is quadratic in the corresponding amplitude, the locked power-response comparison coordinate is

$$\mathcal{R}_T(a,k) \equiv \frac{P_\pi(a,k)}{P_0(a,k)} = C_T(a,k)^2. \tag{108}$$

The identity limit is therefore

$$C_T(a,k) = 1 \quad \Longrightarrow \quad \mathcal{R}_T(a,k) = 1 \quad \Longrightarrow \quad P_\pi(a,k) = P_0(a,k). \tag{109}$$

Route B Option 1 locks the squared-amplitude response as the primary comparison mapping. Its validation boundary is deliberately narrower than a completed native spectral-transport claim: it is a geometry-first bridge validation, not a completed CPTG scale-to-spectrum transport implementation, and it uses the squared-amplitude response rather than an f-to-A_s proxy as its physical map [41]. The bridge lock fixes the amplitude-to-power response relation; it does not by itself supply an independent closed native scale-to-spectrum transport law for $C_T(a,k)$.

The implementation record for this bridge is given in Section 13.1; those stress metrics do not redefine C_T or $\mathcal{R}_T$.

9.2 SkyPi CMB Map Construction and Legacy Envelope Coordinate

9.2.1 Current SkyPi map operator

The current SkyPi map is constructed from the registered source in Section 7.1, passed through the declared beam and mask-forward measurement response. Its observed-phase-conditioned temperature coefficients can be written schematically as

$$a_{\ell m}^{\mathrm{SkyPi}} = a_{\ell m}^{\mathrm{obs}}\, F_\ell \left(\frac{C_\ell^{\mathrm{SkyPi,forward}}}{C_\ell^{\mathrm{obs}}} \right)^{1/2}, \qquad \ell \geq 2, \tag{110}$$

where $C_\ell^{\mathrm{SkyPi,forward}}$ is the forward-processed comparison power in the same declared measurement coordinate as C_ℓ^{obs}, and F_ℓ denotes a common harmonic filter where one is selected (otherwise $F_\ell = 1$). The observed and SkyPi maps receive the same filter and retained-region selection, with nonpositive-power inputs rejected. The observed harmonics supply the sky phases; the fixed SkyPi forward power supplies the comparison amplitudes. The resulting pixel residual and map correlation are evaluated without fitting a global amplitude or offset for each run. These are comparisons of a fixed spectral construction under the selected observational operator, not claims that native Pi geometry generates observed random phases [42].

9.2.2 Legacy Planck/WMAP map-envelope operator

The original comparison-map closure study used a generic envelope coordinate for model M. For an observed temperature field with harmonic coefficients $a_{\ell m}^{\mathrm{obs}}$ and observed angular spectrum C_ℓ^{obs}, its historical envelope is defined by

$$\mathcal{E}_\ell^M \equiv \left(\frac{C_\ell^M}{C_\ell^{\mathrm{obs}}} \right)^{1/2}, \qquad \ell \geq 2. \tag{111}$$

The corresponding phase-locked harmonic comparison coefficients are

$$a_{\ell m}^M = a_{\ell m}^{\mathrm{obs}}\, \mathcal{E}_\ell^M, \tag{112}$$

and the real-space comparison map is

$$T_M(\hat{n}) = \sum_{\ell m} a_{\ell m}^M Y_{\ell m}(\hat{n}). \tag{113}$$

In this legacy operator, the phase scaffold comes from the observed sky, while the original CPTG or control model supplies the power envelope. Its observational conditioning is explicit; it is preserved for the historical tests in [44] and does not supersede the current SkyPi operator.

The historical map-space validation permitted a global amplitude and offset,

$$T_{M,\mathrm{fit}}(\hat{n}) = A_M \widetilde{T}_M(\hat{n}) + B_M, \tag{114}$$

with residual fraction

$$R_M \equiv \frac{\mathrm{RMS}[T_{\mathrm{obs}}(\hat{n}) - T_{M,\mathrm{fit}}(\hat{n})]}{\mathrm{RMS}[T_{\mathrm{obs}}(\hat{n})]}, \tag{115}$$

and relative comparison gap

$$\Delta R \equiv R_{\mathrm{CPTG}} - R_{\mathrm{Planck}}. \tag{116}$$

Here A_M and B_M belong solely to the *legacy* external comparison fit; R_M and ΔR are its downstream historical diagnostics. SkyPi does not perform this per-run amplitude/offset fit. None of these legacy fit quantities is a CPTG-native law, gate, or structural parameter.

The legacy implementation and null-envelope results are recorded in Section 13.2. They retain their original test definitions and are not replacements for SkyPi results.

The legacy study is a real-map comparison-coordinate closure test using a declared observed phase scaffold. It does not imply a native calculation of individual observed CMB phases. Its fitted residual comparisons, smoothing, mask, sky-selection, component-separation, and beam-response conventions are historical and must not be pooled with current SkyPi residuals under different operators [44, 42].

10 Late-Time Growth and Lensing Construction

The spectral-response definitions introduced at the recombination-scale transition are carried forward into the post-recombination structure-growth regime. The following quantities are therefore organized here by their late-time transport role rather than by the observational instrument that measures them.

For the reduced comparison layer used here, the active scale-averaged growth and lensing responses are written as fixed compressed-response forms built from the branch-derived A_L^{CPTG} response scale. These relations organize the comparison calculation; they are not presented here as a completed native scale-transport derivation:

$$\mu_{\mathrm{CPTG}}(a) = 1 + \frac{\pi}{2}\left(A_L^{\mathrm{CPTG}} - 1\right)\left(1 - a^{p_{\mathrm{ac}}}\right), \tag{117}$$

$$\Sigma_{\mathrm{CPTG}}(a) = 1 + \left(A_L^{\mathrm{CPTG}} - 1\right)\left(1 - a^{p_{\mathrm{ac}}}\right). \tag{118}$$

The branch-derived lensing response scale is fixed by the curvature-to-acoustic exponent ratio:

$$A_L^{\mathrm{CPTG}} = \left(\frac{p_C}{p_{\mathrm{ac}}}\right)^2. \tag{119}$$

With

$$p_C = \pi \tag{120}$$

and

$$p_{\mathrm{ac}} = 3 - \frac{\pi}{100}, \tag{121}$$

this gives

$$A_L^{\mathrm{CPTG}} = 1.119956202138. \tag{122}$$

The raw CMB-lensing bandpower comparison is not the same object as the native lensing response. Its registered projection prescription gives the bandpower amplitude

$$A_{\rm bandpower} = 1 + q_{\rm bandpower}\left(A_L^{\rm CPTG} - 1\right), \tag{123}$$

where

$$q_{\rm bandpower} = \frac{1 - \mathcal{G}_T}{\mathcal{B}_\pi}. \tag{124}$$

Therefore,

$$A_{\rm bandpower} = 1.009690662776. \tag{125}$$

The late-time galaxy-lensing amplitude is a transported comparison coordinate rather than an additional native invariant. The locked construction uses the frozen Planck-derived comparison normalization

$$S_{8,\rm anchor} = 0.834,$$

and applies the CPTG geometric transport factor,

$$S_{8,\rm late}^{\rm CPTG} = \mathcal{G}_T S_{8,\rm anchor} = 0.788071335232. \tag{126}$$

The anchor is fixed before the late-lensing comparison and is not re-estimated against DES, HSC, or KiDS data [2].

The compressed CMB-lensing amplitude channel is

$$S_{8,\rm CMBL}^{\rm CPTG} = S_{8,\rm late}^{\rm CPTG}\left(A_L^{\rm CPTG}\right)^{\alpha_C}. \tag{127}$$

The registered compression exponent is

$$\alpha_C = 1 - \frac{2}{\pi}. \tag{128}$$

This gives

$$S_{8,\rm CMBL}^{\rm CPTG} = 0.821191065949. \tag{129}$$

Thus CPTG distinguishes three lensing-related quantities:

$$A_L^{\rm CPTG}, \qquad A_{\rm bandpower}, \qquad S_{8,\rm CMBL}^{\rm CPTG}. \tag{130}$$

The first is the branch-derived response scale, the second is a projected raw bandpower comparison, and the third is a compressed amplitude comparison. These are different observables and should not be collapsed into a single fitted lensing parameter.

The observational weak-lensing and CMB-lensing comparisons for these coordinates are collected in the late-time verification record, Section 15.1.

11 Large-Scale Distance, Horizon, and Acoustic-Ruler Comparison Layer

Before observational conversion, the relevant background relations are fixed by the native pi branch. The late and transport weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \qquad \mathcal{T}_\pi = \frac{1}{\pi}, \tag{131}$$

and the native expansion function and its exact acoustic projection are

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \tag{132}$$

$$E_{\rm ac}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\rm ac}}. \tag{133}$$

Equivalently, $E_{\rm ac}$ is generated from E_π by the projector defined in the core conventions; it is not a second native background.

The acceleration anchor fixes the asymptotic rate and curvature scale through

$$H_\infty = \frac{a_{\pi,0}}{c} = 57.3131992073\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{134}$$

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{135}$$

$$R_{C0} = 48\pi\frac{c^2}{a_{\pi,0}} = 788.7823091477\ \mathrm{Gpc}. \tag{136}$$

The curvature radius evolves on the native branch according to

$$R_C(a) = \frac{R_{C0}}{E_\pi(a)}, \qquad H_\pi(a)R_C(a) = H_0^{(\pi)}R_{C0}. \tag{137}$$

with $H_\pi(a) = H_0^{(\pi)}E_\pi(a)$. This is the curvature-radius evolution used by the companion CPTG cosmological-horizon analysis [40]; it is a native geometric relation and is not a GR/FLRW horizon substitution.

Hubble-like rates are transported by $\sqrt{\mathcal{G}_T}$, giving

$$H_{0,\rm ac}^{\rm CPTG} = H_0^{(\pi)}\sqrt{\mathcal{G}_T} = 67.4777967351\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{138}$$

$$H_{\infty,\rm ac}^{\rm CPTG} = H_\infty\sqrt{\mathcal{G}_T} = 55.7127237837\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{139}$$

The acoustic comparison weights follow directly from the acoustic exponent:

$$\Omega_{m,\rm ac} = \frac{p_{\rm ac}}{3\pi} \simeq 0.314976552850, \tag{140}$$

$$\Omega_{\Lambda,\rm ac} = 1 - \frac{p_{\rm ac}}{3\pi} \simeq 0.685023447150. \tag{141}$$

These relations establish a single analytic source for the displayed decimals.

11.1 Acoustic-Branch Clock and Density-Weighted Diagnostic

The acoustic branch is the exact projection of the native π branch defined above. Its self-synchronized clock uses the acoustic projected rate

$$E_{\rm ac}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\rm ac}}, \tag{142}$$

with $H_{\rm ac}(a) = H_{0,\rm ac}^{\rm CPTG}E_{\rm ac}(a)$ and present age

$$t_{0,\rm ac}^{\rm CPTG} = \frac{1}{H_{0,\rm ac}^{\rm CPTG}}\int_0^1 \frac{da}{aE_{\rm ac}(a)} = 13.88517358\ \mathrm{Gyr}. \tag{143}$$

This is the acoustic branch time coordinate, not a universal clock imposed on the native or luminosity branches.

All dimensional ages use $1\,\mathrm{Mpc} = 3.0856775814913673\times10^{19}\,\mathrm{km}$, $1\,\mathrm{yr} = 31{,}557{,}600\,\mathrm{s}$, and $1\,\mathrm{Gyr} = 3.15576\times10^{16}\,\mathrm{s}$.

The *acoustic-density* object is only a CMB density-weighted diagnostic using $\Omega_{m,\rm ac}$ and $\Omega_{\Lambda,\rm ac}$; it does not introduce a CPTG dark-matter sector and remains distinct from the exact projector weights $\mathcal{T}_\pi$ and $\mathcal{B}_\pi$. Thus

$$E_{{\rm ac},\rho}(a)^2 = \Omega_{\Lambda,\rm ac} + \Omega_{m,\rm ac}a^{-p_{\rm ac}}. \tag{144}$$

$E_{\mathrm{ac},\rho}$ is not another branch or clock and does not replace E_{ac}. Its age-like diagnostic is

$$t_{0,\mathrm{ac},\rho}^{\mathrm{CPTG}} = \frac{1}{H_{0,\mathrm{ac}}^{\mathrm{CPTG}}} \int_0^1 \frac{da}{aE_{\mathrm{ac},\rho}(a)} \tag{145}$$

$$= \frac{2}{p_{\mathrm{ac}} H_{0,\mathrm{ac}}^{\mathrm{CPTG}} \sqrt{\Omega_{\Lambda,\mathrm{ac}}}} \sinh^{-1} \sqrt{\frac{\Omega_{\Lambda,\mathrm{ac}}}{\Omega_{m,\mathrm{ac}}}} \tag{146}$$

$$= 13.926455176722689 \text{ Gyr}. \tag{147}$$

At six-decimal precision this is 13.926455 Gyr; the same p_{ac}-weighted overlay appears in the compressed growth/RSD implementation [51].

11.2 Luminosity Projection, Clock, and Endpoint

The luminosity comparison is constructed theory first and compared with local distance-ladder data only afterward. The parent CPTG polarization homogeneity is $p_{\mathrm{pol}} = 1/3$ [38]. For this comparison-coordinate construction, register the exterior-safe reduced completion

$$\mathcal{L}_\Pi = \beta_{\mathrm{HG}} a_\star^2 F_p(\mathfrak{K}), \qquad F_p(\mathfrak{K}) = \frac{\mathfrak{K}^{p_{\mathrm{pol}}+1}}{1+\mathfrak{K}}. \tag{148}$$

Within the declared reduced completion, choose a unit timelike structural frame

$$U^\mu U_\mu = -1, \qquad h_{\mu\nu} = g_{\mu\nu} + U_\mu U_\nu, \tag{149}$$

and define its physical four-acceleration and projected acceleration-gradient tensor by

$$A_\mu = c^2 U^\nu \nabla_\nu U_\mu, \tag{150}$$

$$B_{\mu\nu} = h_\mu{}^\alpha h_\nu{}^\beta \nabla_\alpha A_\beta. \tag{151}$$

The structural frame defines this reduced comparison invariant; no independent field equation or unique dynamical selection law for U^μ is asserted. The pure-gradient coordinate is

$$\mathfrak{K} \equiv \frac{\kappa^2 B_{\mu\nu} B^{\mu\nu}}{a_\star^2}, \tag{152}$$

Here β_{HG} and κ are not fitted to the luminosity endpoint; β_{HG} cancels from the normalized response, and the matched $I/9$ coefficient is independent of the numerical κ once $\mathfrak{K}_\star = 1/2$, although its physical location may depend on κ. Define

$$W(\mathfrak{K}) \equiv \frac{\mathfrak{K}}{1+\mathfrak{K}}. \tag{153}$$

Then

$$\frac{d\ln F_p}{d\ln \mathfrak{K}} = p_{\mathrm{pol}} + 1 - W. \tag{154}$$

On the reduced gradient branch $\mathfrak{K} \propto a_\star^{-2}$, so the structural-scale derivative is

$$\frac{\partial \ln \mathcal{L}_\Pi}{\partial \ln a_\star} = 2 - 2\frac{d\ln F_p}{d\ln \mathfrak{K}}$$

$$= 2\left(W - p_{\mathrm{pol}}\right). \tag{155}$$

The finite matched stationary condition is therefore

$$W_\star = p_{\mathrm{pol}} = \frac{1}{3}, \tag{156}$$

which gives

$$\mathfrak{K}_\star = \frac{p_{\rm pol}}{1-p_{\rm pol}} = \frac{1}{2}. \tag{157}$$

This finite stationary point is not a theorem of complete covariant dynamical stability; for the polarization sector alone,

$$\left.\frac{\partial^2 \mathcal{L}_\Pi}{\partial(\ln a_\star)^2}\right|_\star = -4p_{\rm pol}(1-p_{\rm pol})\mathcal{L}_\Pi = -\frac{8}{9}\mathcal{L}_\Pi. \tag{158}$$

Full stabilization is not assumed here.

Write the high-gradient carrier as

$$\mathcal{L}_{\rm HG} \equiv \beta_{\rm HG} a_\star^2 \mathfrak{K}^{p_{\rm pol}}, \qquad \mathcal{L}_\Pi = \mathcal{L}_{\rm HG} W, \tag{159}$$

and define the unit-trace spatial metric-response direction

$$M^i{}_j \equiv \frac{B^{ik}B_{jk} + B^{ki}B_{kj}}{2B_{\alpha\beta}B^{\alpha\beta}}, \qquad \operatorname{tr} M = 1. \tag{160}$$

The symmetrized numerator is the exact symmetric inverse-metric response; on the static Hessian branch $B_{ij} = B_{ji}$, it reduces to $M^i{}_j = B^{ik}B_{jk}/(B_{\alpha\beta}B^{\alpha\beta})$.

Below, $\partial/\partial h^{kj}$ is a constitutive partial variation at fixed lower-index $B_{\mu\nu}$, structural frame, $a_\star$, A_μ, $\beta_{\rm HG}$, κ, and $p_{\rm pol}$; only the inverse-metric carrier contraction, hence $\mathfrak{K}$ and W, varies. It is not an unrestricted functional metric variation.

The carrier-normalized inverse-spatial-metric response is

$$\mathsf{C}^i_{\Pi j} \equiv \frac{1}{2\mathcal{L}_{\rm HG}} h^{ik}\frac{\partial \mathcal{L}_\Pi}{\partial h^{kj}} = W\left(p_{\rm pol} + 1 - W\right) M^i{}_j. \tag{161}$$

The overall factor $\beta_{\rm HG}$ cancels from this normalized response. At the matched state,

$$\mathsf{C}_{\Pi,\star} = p_{\rm pol} M. \tag{162}$$

For the isotropic three-dimensional comparison state,

$$M^i{}_j = \frac{1}{3}\delta^i{}_j, \tag{163}$$

and therefore

$$\mathsf{C}_{\Pi,\star} = \frac{1}{9} I. \tag{164}$$

The same rank structure is registered as exact coefficient identities in the acoustic and structural comparison layers. With

$$\ell_\star \equiv \frac{c^2}{a_\star}, \qquad \tau_\star \equiv \frac{c}{a_\star}, \qquad C_R \equiv \frac{R_c}{\ell_\star} = 48\pi, \qquad C_H \equiv H_{\rm struct}\tau_\star = \frac{16\pi}{3}, \tag{165}$$

11.2.1 Three CPTG Derivations of the One-Ninth Luminosity Increment

The matched carrier calculation above gives the first route directly:

$$\mathsf{C}_{\Pi,\star} = \frac{1}{9} I. \tag{166}$$

Its scalar increment is therefore $p_{\rm pol}/3 = (1/3)/3 = 1/9$ in the isotropic three-dimensional matched state.

A second route follows from the separately registered structural scale coefficients. Their ratio is

$$\frac{C_H}{C_R} = \frac{16\pi/3}{48\pi} = \frac{1}{9}. \tag{167}$$

The third route is supplied by the acoustic branch itself. Because

$$\Omega_{m,\mathrm{ac}} = \frac{p_{\mathrm{ac}}}{3\pi}, \qquad \mathcal{G}_T = \frac{p_{\mathrm{ac}}}{\pi}, \tag{168}$$

the common acoustic factor p_{ac}/π cancels exactly and defines the acoustic rank quotient

$$\mathcal{R}_3 \equiv \frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T} = \frac{p_{\mathrm{ac}}/(3\pi)}{p_{\mathrm{ac}}/\pi} = \frac{1}{3}. \tag{169}$$

The acoustic quotient is not merely a numerical route to the same decimal. It is exactly the one-third rank shared by the matched polarization and structural constructions:

$$\mathcal{R}_3 = \frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T} = p_{\mathrm{pol}} = \sqrt{\frac{C_H}{C_R}} = \frac{1}{3}. \tag{170}$$

In the isotropic three-dimensional matched state, the response-direction tensor likewise has eigenvalue one third,

$$M = \mathcal{R}_3 I. \tag{171}$$

Consequently the matched carrier response is

$$\mathsf{C}_{\Pi,\star} = p_{\mathrm{pol}} M = \mathcal{R}_3^2 I, \tag{172}$$

so the square of the acoustic rank quotient is fixed by the same two one-third factors already present in the matched response; it is not an added fitting operation. Therefore

$$\mathcal{R}_3^2 = \left(\frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T}\right)^2 = \frac{C_H}{C_R} = \frac{p_{\mathrm{pol}}}{3} = \frac{1}{9}. \tag{173}$$

The three theory-side constructions therefore close on the same dimensionless increment,

$$\frac{p_{\mathrm{pol}}}{3} = \frac{C_H}{C_R} = \mathcal{R}_3^2 = \left(\frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T}\right)^2 = \frac{1}{9}. \tag{174}$$

These are distinct internal routes—matched response, structural coefficient ratio, and acoustic rank—linked by Eq. (170). They are not statistically independent measurements, do not identify R_c with R_C, and contain no SH0ES input; the external test follows afterward.

At the comparison-coordinate level, the registered affine carrier-plus-response map is

$$\mathsf{Q}_L^{\mathrm{aff}} \equiv I + \mathsf{C}_\Pi. \tag{175}$$

This affine map is a comparison-coordinate prescription, not a universal physical-metric rescaling or unique microscopic luminosity operator.

At the matched isotropic state,

$$\mathsf{Q}_{L,\star}^{\mathrm{aff}} = \left(1 + \frac{1}{9}\right) I = \frac{10}{9} I. \tag{176}$$

The registered affine luminosity normalization is therefore

$$\mathcal{G}_L^{\mathrm{aff}} = 1 + \mathcal{R}_3^2 = 1 + \left(\frac{\Omega_{m,\mathrm{ac}}}{\mathcal{G}_T}\right)^2 = 1 + \frac{1}{9} = \frac{10}{9}. \tag{177}$$

For an isotropic affine co-metric

$$\mathsf{Q}^{\mathrm{aff}}_{L,\star} = q_L I, \qquad q_L = \mathcal{G}^{\mathrm{aff}}_L, \tag{178}$$

the inverse comparison metric has eigenvalue q_L^{-1}. A spatial length or radius coordinate therefore carries the factor $q_L^{-1/2}$. Define

$$R^{\mathrm{eff,aff}}_{C,L}(a) \equiv q_L^{-1/2} R_C(a) = \sqrt{\frac{9}{10}}\, R_C(a). \tag{179}$$

Using the native rate-radius relation in Eq. (137), the reciprocal rate coordinate is

$$\frac{H^{\mathrm{aff}}_L(a)}{H_\pi(a)} = \frac{R_C(a)}{R^{\mathrm{eff,aff}}_{C,L}(a)} = q_L^{1/2} = \sqrt{\frac{10}{9}}. \tag{180}$$

Thus the corresponding rate and reciprocal factors are

$$\Gamma^{\mathrm{aff}}_L = \sqrt{\mathcal{G}^{\mathrm{aff}}_L} = \sqrt{\frac{10}{9}} = 1.054092553389460, \tag{181}$$

$$\Xi^{\mathrm{aff}}_L = \left(\Gamma^{\mathrm{aff}}_L\right)^{-1} = \sqrt{\frac{9}{10}} = 0.948683298050514. \tag{182}$$

The native rate-radius product is preserved identically,

$$H^{\mathrm{aff}}_L(a) R^{\mathrm{eff,aff}}_{C,L}(a) = H_\pi(a) R_C(a). \tag{183}$$

This preservation is a downstream consistency consequence, not an independent derivation of 10/9.

The matched-normalization luminosity comparison trajectory is defined at the same native phase by

$$H^{\mathrm{aff}}_L(a) = \Gamma^{\mathrm{aff}}_L H_\pi(a) = \sqrt{\frac{10}{9}}\, H_\pi(a). \tag{184}$$

At the normalized present phase,

$$H^{\mathrm{aff}}_{0,L} = H_0^{(\pi)} \sqrt{\frac{10}{9}} = 73.1711530416\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{185}$$

Because the same constant multiplies both the full rate history and its present normalization,

$$E^{\mathrm{aff}}_L(a) \equiv \frac{H^{\mathrm{aff}}_L(a)}{H^{\mathrm{aff}}_{0,L}} = \frac{H_\pi(a)}{H_0^{(\pi)}} = E_\pi(a). \tag{186}$$

The equality is exact: the normalized shape and native phase remain unchanged, while only the comparison normalization changes. "Matched" does not assert $\mathfrak{K} = 1/2$ in every microscopic environment or epoch.

In general, a luminosity comparison projector may be written

$$H_{\mathrm{lum}}(N_\pi) = \Gamma_L(N_\pi) H_\pi(N_\pi), \qquad \Xi_{\mathrm{lum}}(N_\pi) \equiv \Gamma_L(N_\pi)^{-1}. \tag{187}$$

For the affine matched-normalization trajectory, $\Gamma_L(N_\pi) = \Gamma^{\mathrm{aff}}_L$ is constant and the branch clocks satisfy

$$\frac{dt^{\mathrm{aff}}_L}{dt_\pi} = \Xi^{\mathrm{aff}}_L = \sqrt{\frac{9}{10}}, \qquad H^{\mathrm{aff}}_L\, dt^{\mathrm{aff}}_L = H_\pi\, dt_\pi = dN_\pi. \tag{188}$$

With the same zero-time convention,

$$t^{\mathrm{aff}}_L(a) = \sqrt{\frac{9}{10}}\, t_\pi(a), \tag{189}$$

and the present comparison-clock coordinate is

$$t_{0,L}^{\rm aff} = 12.0996216046 \text{ Gyr}. \tag{190}$$

This is a luminosity comparison-clock coordinate, not a second physical age of the Universe.

The four-anchor SH0ES distance-ladder value [4]

$$H_0^{\rm SH0ES} = 73.17 \pm 0.86 \text{ km s}^{-1}\,\text{Mpc}^{-1} \tag{191}$$

is therefore used as an independent external observational validation of the already-derived luminosity normalization rather than as its defining input. Its native-relative rate coordinate is

$$\Gamma_{\rm loc}^{\rm SH0ES} = \frac{H_0^{\rm SH0ES}}{H_0^{(\pi)}} = 1.054075942846, \tag{192}$$

$$\sigma_\Gamma = 0.012389029805, \tag{193}$$

so

$$\Gamma_{\rm loc}^{\rm SH0ES} = 1.054075942846 \pm 0.012389029805. \tag{194}$$

The theory-constructed affine endpoint differs from the SH0ES central value by

$$H_{0,L}^{\rm aff} - H_0^{\rm SH0ES} = 0.0011530416 \text{ km s}^{-1}\,\text{Mpc}^{-1}, \tag{195}$$

or approximately 0.00134σ using the declared SH0ES uncertainty. This agreement is the independent external observational validation of the theory-side 10/9 normalization. It is not used in the matched-carrier, structural-coefficient, or acoustic-rank derivations in Eqs. (164), (167), and (173), nor is it used to define the affine endpoint in Eq. (185).

For continuity with the Hubble-bridge and horizon analyses [39, 40], the historical SH0ES-conditioned luminosity scaffold is retained as a separate comparison object:

$$H_{\rm lum}^{\rm anchor}(a) = H_{0,\rm lum}^{\rm anchor} E_{{\rm ac},\rho}(a), \qquad H_{0,\rm lum}^{\rm anchor} \equiv H_0^{\rm SH0ES}. \tag{196}$$

Its phase-dependent clock map is

$$\Xi_{\rm lum}^{\rm anchor}(a) = \frac{H_\pi(a)}{H_{\rm lum}^{\rm anchor}(a)} \tag{197}$$

$$= \frac{H_0^{(\pi)}}{H_{0,\rm lum}^{\rm anchor}} \left[\frac{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}{\Omega_{\Lambda,\rm ac} + \Omega_{m,\rm ac} a^{-p_{\rm ac}}}\right]^{1/2}. \tag{198}$$

This gives

$$t_{0,\rm lum}^{\rm anchor} = \int_0^1 \frac{da}{a H_{\rm lum}^{\rm anchor}(a)} = 12.8430574232 \text{ Gyr}. \tag{199}$$

The affine trajectory and the SH0ES-conditioned scaffold answer different questions. Their present endpoints are nearly identical, but their histories are not equal because $E_L^{\rm aff} = E_\pi$ whereas the historical scaffold uses $E_{{\rm ac},\rho}$. The scaffold remains data-conditioned and is not a second branch or a microscopic luminosity projector.

The theory-constructed acoustic-to-luminosity endpoint separation is

$$\frac{H_{0,L}^{\rm aff}}{H_{0,\rm ac}^{\rm CPTG}} - 1 = 0.0843737730 = 8.43737730\%. \tag{200}$$

11.3 Acoustic–Luminosity Midpoint Comparison Coordinate

Some CPTG sensitivity calculations require a central comparison coordinate between the acoustic and luminosity endpoints without promoting that central value to another cosmological branch. The registered symmetric midpoint prescription remains

$$\omega_{\mathrm{AL}} \equiv \frac{1}{2}. \tag{201}$$

For any two endpoint quantities $Q_{\mathrm{ac}}(N_\pi)$ and $Q_{\mathrm{lum}}(N_\pi)$ that have first been evaluated in the same comparison coordinate at the same native phase, the midpoint readout is

$$Q_{\mathrm{mid}}(N_\pi) \equiv (1-\omega_{\mathrm{AL}})Q_{\mathrm{ac}}(N_\pi) + \omega_{\mathrm{AL}}Q_{\mathrm{lum}}(N_\pi). \tag{202}$$

The weight $\omega_{\mathrm{AL}} = 1/2$ is a registered comparison rule, not another native invariant, branch exponent, or fitted parameter.

Using the exact acoustic comparison and the affine matched-normalization luminosity coordinate, the primary rate-space midpoint is

$$H^{\mathrm{aff}}_{\mathrm{mid}}(a) \equiv \frac{1}{2}\left[H_{\mathrm{ac}}(a) + H^{\mathrm{aff}}_{L}(a)\right]. \tag{203}$$

Its native-relative rate coordinate is

$$\mathcal{M}^{\mathrm{aff}}_{\mathrm{AL}}(a) \equiv \frac{H^{\mathrm{aff}}_{\mathrm{mid}}(a)}{H_\pi(a)}. \tag{204}$$

At the present phase,

$$\begin{aligned}\mathcal{M}^{\mathrm{aff}}_{\mathrm{AL},0} &= \frac{1}{2}\left[\sqrt{\mathcal{G}_T} + \sqrt{\frac{10}{9}}\right] \\ &= 1.013083738870,\end{aligned} \tag{205}$$

and

$$\begin{aligned}H^{\mathrm{aff}}_{0,\mathrm{mid}} &= H_0^{(\pi)}\mathcal{M}^{\mathrm{aff}}_{\mathrm{AL},0} \\ &= 70.3244748884\ \mathrm{km\,s^{-1}\,Mpc^{-1}}.\end{aligned} \tag{206}$$

This theory-constructed midpoint inherits the status of the exact acoustic projection and the registered affine luminosity comparison prescription. It remains a comparison readout rather than a third branch.

For continuity with earlier sensitivity calculations, applying the same midpoint rule to the SH0ES-conditioned anchor endpoint gives

$$\mathcal{M}^{\mathrm{anchor}}_{\mathrm{AL},0} = 1.013075433598 \pm 0.006194514903, \tag{207}$$

and

$$H^{\mathrm{anchor}}_{0,\mathrm{mid}} = 70.3238983675 \pm 0.43\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{208}$$

These anchor midpoint values remain data-conditioned and must not be conflated with the affine matched-normalization midpoint.

For nonlinear observables, midpoint formation is performed after each endpoint observable has been evaluated. For example, if the two endpoint distance coordinates are $D_{\mathrm{ac}}(N_\pi)$ and $D_{\mathrm{lum}}(N_\pi)$, then

$$D_{\mathrm{mid}}(N_\pi) = (1-\omega_{\mathrm{AL}})D_{\mathrm{ac}}(N_\pi) + \omega_{\mathrm{AL}}D_{\mathrm{lum}}(N_\pi), \tag{209}$$

rather than constructing a third branch rate and then re-integrating it as though it were an independent cosmology.

11.4 Common-Phase Horizon and Last-Scattering Coordinates

The branch clocks are compared only at equal geometric phase. For the Planck last-scattering reference

$$z_\star = 1089.92, \qquad a_{\rm dec} = \frac{1}{1+z_\star} = 9.1665750009 \times 10^{-4}, \tag{210}$$

the CPTG acoustic comparison distance is

$$\chi_\star = \frac{c}{H_0^{(\pi)}} \int_0^{z_\star} \frac{dz}{E_{\rm ac}(z)} = 13876.8219 \text{ Mpc}. \tag{211}$$

The dimensional prefactor is the native normalization $H_0^{(\pi)}$; the exact acoustic projection enters through $E_{\rm ac}$. This convention is important because substituting $H_{0,\rm ac}^{\rm CPTG}$ into the prefactor would define a different distance coordinate.

The physical radius of the region that becomes the observed last-scattering sphere is

$$L_\star(a) = a\chi_\star. \tag{212}$$

At decoupling,

$$L_\star(z_\star) = 12.72029287 \text{ Mpc}, \qquad R_C(z_\star) = 23.64711498 \text{ Mpc}, \tag{213}$$

and therefore

$$\frac{L_\star(z_\star)}{R_C(z_\star)} = 0.53792156 < 1. \tag{214}$$

This is a radius-to-radius comparison at one geometric phase. A diameter-to-radius ratio is not used as a containment criterion [40].

The equality $L_\star = R_C$ occurs at

$$z_{\rm enter} = 3231.635355, \qquad L_\star = R_C = 4.292727257 \text{ Mpc}, \tag{215}$$

with native thermal temperature

$$T_{\rm enter} = T_0(1+z_{\rm enter}) = 8810.483007 \text{ K}. \tag{216}$$

This entry phase marks geometric containment; it is not defined as the beginning of coherence.

The native and exact acoustic clocks at the same two geometric phases are

$$z_{\rm enter}: \qquad t_\pi \simeq 48{,}802 \text{ yr}, \qquad t_{\rm ac} \simeq 106{,}889 \text{ yr}, \tag{217}$$

$$z_\star: \qquad t_\pi \simeq 268{,}835 \text{ yr}, \qquad t_{\rm ac} \simeq 536{,}006 \text{ yr}. \tag{218}$$

These are branch-specific integrations at common N_π, not a cross-branch synchronization condition. Together with Eqs. (187)–(188), they complete the reference architecture of one native branch and two projected branch clocks.

Using the Planck acoustic-angle anchor $100\theta_\star = 1.04110$, the distance above corresponds arithmetically to

$$r_{\rm ac} = \theta_\star \chi_\star = 144.4716 \text{ Mpc}, \tag{219}$$

which is close to the Planck-inferred $r_\star = 144.43 \pm 0.26$ Mpc [2]. This is a comparison-coordinate reconstruction from the adopted angular anchor, not an independent prediction of $\theta_\star$.

For a spatially flat comparison branch, the transverse comoving distance is

$$D_M(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}, \tag{220}$$

the radial Hubble distance is

$$D_H(z) = \frac{c}{H_0 E(z)}, \tag{221}$$

and the volume-averaged acoustic distance is

$$D_V(z) = \left[z D_M(z)^2 D_H(z) \right]^{1/3}. \tag{222}$$

BAO likelihoods compare acoustic ratios rather than dimensional distances alone. The primary coordinates are

$$\frac{D_M(z)}{r_d}, \qquad \frac{D_H(z)}{r_d}, \qquad \frac{D_V(z)}{r_d}. \tag{223}$$

For anisotropic BAO or ShapeFit-style comparisons,

$$q_\parallel = \frac{D_H(z)/r_d}{[D_H(z)/r_d]_{\rm fid}}, \tag{224}$$

$$q_\perp = \frac{D_M(z)/r_d}{[D_M(z)/r_d]_{\rm fid}}, \tag{225}$$

$$q_{\rm iso} = (q_\perp^2 q_\parallel)^{1/3}, \tag{226}$$

$$q_{\rm ap} = \frac{q_\parallel}{q_\perp}. \tag{227}$$

When an absolute BAO-ruler normalization is required by the present compressed implementation, the fixed working comparison normalization is

$$r_d = 148.998491 \text{ Mpc}. \tag{228}$$

In the existing compressed BAO analysis it is an externally fitted comparison-layer normalization, not a native geometric invariant. The separate quarter-ruler relation is dimensionless and is not used to reclassify this externally fitted absolute normalization as a native quantity.

Supernova comparisons use

$$D_L(z) = (1+z) D_M(z), \qquad \mu(z) = 5 \log_{10} \left[\frac{D_L(z)}{\text{Mpc}} \right] + 25. \tag{229}$$

The supernova comparison is therefore a distance-shape comparison after standardization, intercept treatment, calibration handling, and covariance weighting [5]. BAO comparisons are made through survey-defined distance/ruler ratios and AP coordinates [7, 8, 6]. In both cases CPTG supplies the geometric distance relation first; the observational pipeline determines which comparison coordinates are evaluated.

12 Primordial Network Implementation and Verification Record

The microscopic and primordial comparison layer is tested after its coordinates are fixed. The following validations therefore remain downstream of the source, reaction-observable, network-response, and mass-seven definitions above; they do not define those coordinates.

12.1 Commissioning-Channel Reaction-Observable Verification

The fixed zero-energy commissioning normalization is compared with the Solar Fusion II LUNA-extrapolated value $S_{12}(0) = 0.216 \pm 0.010\,\text{eV b}$, giving a signed residual of -0.149960σ [16, 15]. At $E_0 = 6.64\,\text{keV}$, the fixed finite-energy coordinate is compared with $0.252 \pm 0.018\,\text{eV b}$, or -0.020433σ [16, 15].

The later universal-reaction audit provides a direct empirical check of this observable layer without changing its normalization. The complete Mossa/LUNA replay places all 13 published S-factor points from 32.4 to 262.9 keV inside their combined one-standard-deviation intervals and all 17 tabulated thermonuclear rates from $T_9 = 0.001$ to 10 inside their published asymmetric one-standard-deviation bounds [17, 22]. The same fixed projection is consistent with the Tišma and Turkat covariance-weighted shape tests, while the three-point Dhibar dataset remains strongly discordant because of its highest-energy point; no theory parameter is changed to absorb that discrepancy [18, 19, 20].

Observable test	CPTG result	External comparison	Classification
Zero-energy $S_{12}(0)$	0.2145003969219307 eV b	0.216 ± 0.010 eV b	-0.149960σ
Solar-Gamow $S_{12}(6.64\,\mathrm{keV})$	0.2516322073981925 eV b	0.252 ± 0.018 eV b	-0.020433σ; external shape
Mossa/LUNA S-factor grid	13/13 inside combined 1σ	32.4–262.9 keV	max. $\lvert z_S \rvert = 0.951713$
Mossa/LUNA rate grid	17/17 inside published interval	$T_9 = 0.001$–10	max. $\lvert \widetilde{z}_R \rvert = 0.959051$
91 keV pivot	0.7931440807 eV b	0.799 ± 0.018 eV b	-0.3253σ
Covariance shape checks	Tišma $\chi^2/\nu = 0.2325$; Turkat 1.1137	fixed projection	consistency diagnostics
Dhibar three-point check	$\chi^2/\nu = 15.5173$	fixed projection	discordant; retained

Table 13: Commissioning-channel reaction-observable comparisons. The zero-energy normalization is the CPTG-derived coordinate. Finite-energy and thermal rows retain their declared external energy-shape information. The Mossa/LUNA interval counts and covariance quantities are validation metrics, not additional theory parameters.

12.2 PRIMAT-Native Baryon, Radiation, and Mass-Seven Validation

PRIMAT v0.3.2 is the primary numerical authority for the present full-coordinate primordial-network test [14]. The completed, prospectively preregistered large-network protocol first calibrates the native radiation response with two gate-off rows. This abundance-independent calibration is a native translation step used to implement the already-fixed CPTG radiation coordinate; it is not an independent PRIMAT prediction of that target. The reported values

$$N_{\mathrm{eff}}(\Delta N_{\mathrm{eff}} = 0) = 3.0439772985579183, \tag{230}$$

$$N_{\mathrm{eff}}(\Delta N_{\mathrm{eff}} = 0.01) = 3.0539772985579177 \tag{231}$$

give

$$\frac{dN_{\mathrm{eff}}}{d(\Delta N_{\mathrm{eff}})} = 0.9999999999999343. \tag{232}$$

The prospectively frozen native input required to implement the already-defined CPTG radiation coordinate is therefore

$$\Delta N_{\mathrm{eff}}^{\mathrm{CPTG}} = -0.07539322509392317, \tag{233}$$

set before target abundance exposure. At the joint baryon-plus-radiation target

$$\left((\Omega_b h^2)_{\mathrm{BBN}}, N_{\mathrm{eff}}^{\mathrm{CPTG}}\right) = (0.021898765370, 2.968584073464)\,, \tag{234}$$

PRIMAT returns

$$N_{\mathrm{eff}}^{\mathrm{PRIMAT}} = 2.968584073463995, \tag{235}$$

an absolute coordinate error of only 4.88×10^{-15}. This residual is a native translation/mapping-closure metric after the frozen abundance-independent translation; it is not an independent PRIMAT prediction of $N_{\mathrm{eff}}^{\mathrm{CPTG}} = p_{\mathrm{ac}}$. No deuterium, helium, or lithium abundance is used to construct the radiation mapping.

The observational background controls are

$$10^6(\mathrm{D/H})_{\mathrm{obs}} = 25.08 \pm 0.29, \qquad Y_{p,\mathrm{obs}} = 0.245 \pm 0.003 \tag{236}$$

[13]. The lithium value is retained only as a downstream diagnostic and is excluded from the scientific acceptance gates.

Quantity	Native PRIMAT result	Frozen reference or control	Residual / role
$N_{\rm eff}$, target	2.968584073463995	2.968584073464	4.88×10^{-15} absolute; pass
D/H, gate off	$25.0669253584 \times 10^{-6}$	$(25.08 \pm 0.29) \times 10^{-6}$	-0.045085σ; pass
Y_p, gate off	0.245760246933	0.245 ± 0.003	$+0.253416\sigma$; pass
Live $A = 7$ survival	0.318366498376680	$1/\pi = 0.318309886183791$	0.0177852% relative; pass
D/H gate disturbance	3.91×10^{-9} relative	$< 10^{-6}$	Selectivity pass
Y_p gate disturbance	3.86×10^{-12} relative	$< 10^{-6}$	Selectivity pass
^{3}He/H gate disturbance	5.59×10^{-10} relative	$< 10^{-6}$	Selectivity pass
^{7}Li/H, gate off	$5.2735647904 \times 10^{-10}$	$(1.45 \pm 0.25) \times 10^{-10}$	$+15.2943\sigma$; diagnostic only
^{7}Li/H, gate on	$1.6789263563 \times 10^{-10}$	$(1.45 \pm 0.25) \times 10^{-10}$	$+0.915705\sigma$; diagnostic only

Table 14: Completed prospectively preregistered PRIMAT native-coordinate implementation and downstream response validation at the CPTG transported baryon and radiation coordinate. D/H and Y_p are preregistered background gates; the mass-seven survival and non-mass-seven disturbance rows test the live transport law; lithium abundance is report-only and does not enter the hard decision [14, 13, 23].

The gate-off deuterium-to-hydrogen endpoint in the preceding completed-coordinate network row is $\mathrm{D/H} = 2.50669253584 \times 10^{-5}$. An earlier commissioned $D(p,\gamma)^3$He study reported $\mathrm{D/H} = 2.5074364788625683 \times 10^{-5}$ for its separately declared calculation [15, 23]. The two values belong to different implementation/test contexts; the commissioning result is retained as legacy context and is not substituted for, combined with, or used to revise the frozen complete-coordinate PRIMAT endpoint. Both are *D/H number ratios*, not free-proton source coordinates.

With the gate enabled, the non-mass-seven native outputs are

$$\mathrm{D/H}_{\rm gated} = 2.5066925456419876 \times 10^{-5}, \tag{237}$$

$$Y_{p,\rm gated} = 0.24576024693245754, \tag{238}$$

$$(^3\mathrm{He/H})_{\rm gated} = 1.0503519364716970 \times 10^{-5}. \tag{239}$$

The archived gate-off helium output $Y_p^{\rm CPTG} = 0.24576024693340703$ is unchanged. Its separate 2026 observational addendum compares this one frozen result with LBT $Y_p = 0.2458 \pm 0.0013$ and EMPRESS XV $Y_p = 0.2402 \pm 0.0040$; those are observational references, not new CPTG inputs or revised network outputs [43]. The same frozen helium value enters the current SkyPi CMB source under Equation (96).

The network-realized live mass-seven survival differs from the exact isolated coordinate $\mathcal{S}_7 = 1/\pi$ because ordinary native production and destruction continue during the transport window; the measured deviation is therefore a genuine network-response test rather than a built-in algebraic identity.

All 23 preregistered scientific gates pass. The result establishes completed prospectively preregistered PRIMAT-native implementation and mapping closure for the jointly transported CPTG baryon and radiation coordinate, together with downstream native-network response and selectivity validation of the locked live $A = 7$ abundance-transport law for the tested PRIMAT v0.3.2 large-network configuration. It does not imply that PRIMAT independently derives $N_{\rm eff}^{\rm CPTG} = p_{\rm ac}$ or $\tau_7 = \ln \pi$, and it does not convert τ_7 into a stellar depletion factor, a laboratory cross section, a post-processing abundance correction, or a fifth coordinate of the four-sector nuclear simplex.

12.3 PRIMAT-Native First-Order Reaction Pushforward Validation

The network-conditioned coordinate in Equation (86) is tested separately from the mass-seven transport law. PRIMAT's completed full-reaction authority contains 20,550/20,550 committed native rows: six baseline rows plus $428 \times 6 \times 8 = 20{,}544$ frozen reaction–anchor–rate-branch rows. Thus

$$6 + 428 \times 6 \times 8 = 20{,}550. \tag{240}$$

A separate native source/Jacobian campaign constructs 84 first-order variational predictions from fourteen mapped light-sector reactions at the same six anchors. The prediction table is frozen before the authoritative rate-varied endpoint coefficients are opened, so the endpoint response is an evaluation target rather than a fitting input [22, 14].

The endpoint reference at each reaction–anchor pair is independently extracted from the completed eight-branch ladder through

$$\Delta \ln Y = c_1 x + c_2 x^2, \qquad x = -\ln f_r, \tag{241}$$

with zero intercept. For a primary resolved first-order response, $\|\mathbf{c}_{1,\mathrm{ref}}\| \geq 10^{-3}$, the endpoint-vector discrepancy is

$$\epsilon_{\mathrm{vec}} = \frac{\|\mathbf{c}_{1,\mathrm{var}} - \mathbf{c}_{1,\mathrm{ref}}\|}{\|\mathbf{c}_{1,\mathrm{ref}}\|}, \tag{242}$$

and the direction metric is the ordinary cosine

$$\cos\theta = \frac{\mathbf{c}_{1,\mathrm{var}} \cdot \mathbf{c}_{1,\mathrm{ref}}}{\|\mathbf{c}_{1,\mathrm{var}}\| \|\mathbf{c}_{1,\mathrm{ref}}\|}. \tag{243}$$

The frozen gates are $\epsilon_{\mathrm{vec}} \leq 0.02$ and $\cos\theta \geq 0.999$. Independent BDF and Radau propagations are also required to agree under the separately frozen norm-based cross-solver relative-difference metric, $\epsilon_{\mathrm{BDF/Radau}} \leq 5 \times 10^{-4}$, with cross-solver direction cosine at least 0.999999.

Validation quantity	PRIMAT result	Interpretation / gate
Frozen source/Jacobian predictions	84/84	Completed before endpoint-reference exposure
Primary resolved tests	48/48 pass	$\|\mathbf{c}_{1,\mathrm{ref}}\| \geq 10^{-3}$
Primary + secondary resolved cases	66/84	Response norm $\geq 10^{-4}$; secondary floor
Worst primary endpoint-vector discrepancy	3.6762871×10^{-4} (0.0367629%)	Frozen requirement ≤ 0.02
Minimum primary endpoint direction cosine	0.9999999921	Frozen requirement ≥ 0.999
Maximum BDF/Radau cross-relative difference	5.8248×10^{-7}	Frozen requirement $\leq 5 \times 10^{-4}$
Maximum source-direction residual	3.17×10^{-16}	Frozen requirement $< 10^{-10}$
Maximum baryon-conservation residual	4.44×10^{-16}	Frozen requirement $< 10^{-10}$

Table 15: Completed PRIMAT first-order native source/Jacobian validation of the network-conditioned nuclear reaction-response pushforward. The authoritative endpoint coefficients are not used to construct the frozen variational predictions [22, 14].

For the commissioned $D(p,\gamma)^3$He channel, the largest six-anchor endpoint-vector discrepancy across either variational solver is approximately 1.62×10^{-4} (0.0162%), with direction cosine effectively unity. This result is especially important for comparison-coordinate interpretation: it validates the route

$$\mathbf{s}_r \longrightarrow \mathbf{d}^{(1)}_{r,\mathcal{N}} \longrightarrow \mathbf{y}^{(1)}_{r,\mathcal{N}} \longrightarrow \mathcal{C}^{(1)}_{r,\mathcal{N}}, \tag{244}$$

not the shortcut $\mathbf{s}_r \equiv \mathbf{c}_1$.

PArthENoPE 3.0 supplies the completed second-authority transfer in the shared endpoint space. Its archived campaign contains 695/695 native rows, 338/338 matched pairs, and 84/84 complete ladders; the declared Reaction-20 transfer has direction cosine 0.999979437501036, with all component-ratio and six-anchor stability gates satisfied [22, 24]. This supports the same comparison architecture while preserving code-local currents, source kernels, Jacobians, normalizations, and solver trajectories as network-specific objects. PRyMordial full-native authority constitutes a separate validation lane and is not required to define the PRIMAT first-order coordinate reported here.

13 CMB Implementation and Full-Stack Verification Record

The definitions in the preceding CMB construction sections remain authoritative. The material below records how those fixed coordinates were exercised numerically; it does not redefine them.

13.1 Curvature-Transport Spectrum Bridge Verification

The bridge was exercised through CAMB and multiple Planck likelihood coordinates. The NPIPE TTTEEE stress ladder is monotonic from the identity row $f_r = 1$ through $f_r = 0.990$, where the accumulated shift is $\Delta\chi^2 = +2.923593$. NPIPE

TT, TE, EE, and TTTEEE split sectors pass monotonically. Plik TT, EE, and TTTEEE also pass monotonically; the TE sector contains a localized diagnostic inversion with its minimum at $f_r = 0.996$ and $\Delta\chi^2 = -0.445906$. In the near-identity full stack including lensing, $f_r = 0.999$ gives $\Delta\chi^2 = +0.350109$. These f_r values are stress coordinates only, and the summed χ^2 values are bridge smoke metrics rather than replacements for the profiled CMB validation reported later in this article [41].

13.2 Legacy CMB Map-Envelope Verification

Under the declared Planck SMICA comparison, the fitted CPTG map has residual RMS $18.218928\,\mu$K and valid-sky correlation 0.983434. Across the five principal Planck component maps, the mean CPTG-minus-Planck full-band residual-RMS gap is only $0.023518\,\mu$K. The SMICA full/half-mission/odd-even split test gives a mean gap of $0.023891\,\mu$K with a split-to-split span of $0.000690\,\mu$K. The null-envelope audit is the uniqueness control: mean residual RMS is $18.230\,\mu$K for the Planck benchmark and $18.253\,\mu$K for the locked CPTG envelope, while shuffled-spectrum nulls rise to $95.461\,\mu$K. Thus phase locking alone is insufficient to generate the observed near-degeneracy; the angular-power envelope must retain the relevant acoustic structure [44].

These original-test statistics remain data-conditioned historical measurements under the legacy fitted map operator; they are not SkyPi results or native predictions of the random CMB phase realization [42].

13.3 Fixed-Row Likelihood Verification Protocol

The cleaned CMB branch is evaluated with fixed CPTG cosmology and fixed comparison gates. The comparison statistic is

$$\Delta\chi^2 = \chi^2_{\mathrm{CPTG,locked}} - \chi^2_{\mathrm{control}}, \tag{245}$$

where the control row is evaluated under the same likelihood, nuisance treatment, profiling or sampling protocol, and data combination.

For this fixed-row robustness audit, the comparison categories are

$$\Delta\chi^2 \le 5 \quad \Longrightarrow \quad \text{control level}, \tag{246}$$

$$5 < \Delta\chi^2 \le 10 \quad \Longrightarrow \quad \text{watch/stress}, \tag{247}$$

and

$$\Delta\chi^2 > 10 \quad \Longrightarrow \quad \text{decisive stress}. \tag{248}$$

These pragmatic audit categories are motivated by established Planck and Plik/CamSpec-NPIPE likelihood-robustness scales and are not parameter-estimation confidence contours or universal model-selection thresholds [3, 37].

All CMB comparisons in this section keep the locked gates fixed:

$$\mathcal{D}_c = 0.9735353159758136, \tag{249}$$

$$\mathcal{A}_s = \frac{\pi}{3}, \tag{250}$$

$$\mathcal{T}_\tau = \mathcal{G}_T^2, \tag{251}$$

and

$$A_{\rm lens} = 1. \tag{252}$$

No result in this section retunes the cold-sector transport, scalar-amplitude gate, reionization-optical-depth gate, or lensing-amplitude setting.

13.4 Validation Milestone Summary

Table 16 summarizes the fixed-row profile validation, diagnostics, posterior stability, and exact gate closure; MCMC stability is not relabeled as conventional convergence.

Validation layer	Result summary	Status
High-ℓ profile robustness	Plik TTTEEE, SPT-3G TEEE, NPIPE CamSpec TTTEEE: mean median shift +2.221794; maximum median shift +4.501564; CamSpec worst repeat +5.013677.	Validation pass; CamSpec watch
Posterior-proxy stability check	Plik TTTEEE and SPT-3G TEEE: mean shift +1.6722; maximum shift +1.9233.	Control-level stability; convergence not claimed
Low-ℓ and lensing diagnostic stability	Planck low-ℓ EE, Planck lensing, ACT DR6 lensing: Planck mean +1.589; maximum Planck shift +2.149.	Diagnostic pass
Full-stack profile validation	Plik+lowE+lensing, NPIPE CamSpec+lowE+lensing, SPT+ACT: mean primary shift +2.101464; maximum primary shift +3.669953; CamSpec worst repeat +5.990520, in the watch/stress band.	Validation pass; CamSpec watch repeat retained
Combined-stack posterior stability	Plik+lowE+lensing; NPIPE CamSpec+lowE+lensing; SPT+ACT: mean median shift +0.096642; maximum median shift +0.737534; maximum reported $R-1 = 0.143463$.	Control-level stability; no convergence claim
Final validation consolidation	Consolidated profile, posterior-stability, and gate status: maximum designated consolidation statistic $\lvert\Delta\chi^2\rvert = 4.824743$ with $A_{\rm lens} = 1$; the separate CamSpec worst-repeat diagnostic is 5.990520.	Profile validation with supporting stability diagnostics; worst repeat in watch band
Amplitude/reionization-depth gate closure	Gates $\mathcal{A}_s = \pi/3$ and $\mathcal{T}_\tau = \mathcal{G}_T^2$: exact closure with maximum relative difference 0.	Numerical validation pass

Table 16: CMB validation milestones for the fixed $\mathcal{G}_T^2$ comparison row.

13.5 High-ℓ Profile Robustness

The fixed comparison row is tested against distinct high-ℓ likelihood implementations using repeated nuisance-profile comparisons. The aggregate designated median shift is

$$\left\langle \Delta\chi^2_{\rm median} \right\rangle = 2.221794, \tag{253}$$

and the maximum designated high-ℓ median shift is

$$\max\left(\Delta\chi^2_{\rm median}\right) = 4.501564. \tag{254}$$

Plik and SPT-3G are control-level; NPIPE CamSpec is also control-level by the designated statistic, with the retained profile sensitivity documented below.

13.6 Plik and SPT Posterior-Proxy Stability

Plik and SPT-3G are also tested with posterior-proxy sampling; because the reported $R-1$ values exceed Cobaya's default 0.01 stopping criterion, this is a likelihood-stability diagnostic rather than a convergence claim [30, 31].

Target	CPTG proxy χ^2	Control proxy χ^2	Shift	CPTG samples	Status
Plik TTTEEE	2344.4088	2342.4855	+1.9233	33520	Control-level
SPT-3G TEEE	1112.9661	1111.5450	+1.4211	18680	Control-level

Table 17: Plik/SPT posterior-proxy likelihood-stability comparison for the fixed comparison row.

The corresponding control chains contain 57600 accepted Plik samples and 28120 accepted SPT samples. The final mean $R-1$ values are

$$R_{\mathrm{Plik,CPTG}} - 1 = 0.029169, \qquad R_{\mathrm{Plik,control}} - 1 = 0.029464, \tag{255}$$

$$R_{\mathrm{SPT,CPTG}} - 1 = 0.028239, \qquad R_{\mathrm{SPT,control}} - 1 = 0.025843. \tag{256}$$

These $R-1$ values are not conventionally converged chains [31].

The mean and maximum designated likelihood shifts are

$$\left\langle \Delta\chi^2_{\mathrm{posterior}} \right\rangle = 1.6722, \qquad \max(\Delta\chi^2_{\mathrm{posterior}}) = 1.9233. \tag{257}$$

The likelihood shifts remain control-level.

13.7 Low-ℓ and Lensing Diagnostic Stability

Table 18 gives the low-ℓ and lensing diagnostics with fixed τ_{reio} and $A_{\mathrm{lens}} = 1$.

Target	CPTG median χ^2	Control median χ^2	Median shift	Status
Planck low-ℓ EE	396.550	395.521	+1.029	Control-level
Planck 2018 lensing	10.936	8.88949	+2.149	Control-level
ACT DR6 lensing	14.7052	13.9677	+0.7375	Control-level

Table 18: Low-ℓ and lensing diagnostic stability for the fixed comparison row.

For the designated Planck low-ℓ and Planck lensing rows, the mean median shift is

$$\left\langle \Delta\chi^2_{\mathrm{lowE/lensing}} \right\rangle = 1.589, \tag{258}$$

and the maximum designated median shift is

$$\max\left(\Delta\chi^2_{\mathrm{lowE/lensing}}\right) = 2.149. \tag{259}$$

The fixed reionization-depth relation

$$\tau_{\mathrm{reio,eff}} = 0.05893087593426084 \tag{260}$$

remains inside the reported diagnostic control region.

13.8 Combined-Stack Profile Validation

The combined-stack profile audit keeps the same fixed CMB row. NPIPE CamSpec is a distinct PR4 likelihood implementation on overlapping Planck sky information, not statistically independent data; when CamSpec-NPIPE-lite is unavailable, full CamSpec uses the stated best-of-N protocol [37].

All enabled combined-stack profile runs succeed:

$$44/44 = 1. \tag{261}$$

There are no timeouts and no run failures. The designated combined-stack results are given in Table 19.

Stack	Protocol	Primary shift	Median shift	Worst shift
Plik TTTEEE + lowE + Planck lensing	Median profile	+1.896906	+1.896906	+2.418011
NPIPE CamSpec + lowE + Planck lensing	Best-of-N profile	+3.669953	+4.824743	+5.990520
SPT-3G TEEE + ACT DR6 lensing	Median profile	+0.737534	+0.737534	+0.737534

Table 19: Locked combined-stack CMB profile validation. NPIPE CamSpec is treated as a distinct PR4 likelihood implementation using full CamSpec best-of-N when CamSpec-NPIPE-lite is unavailable; it is not statistically independent of other Planck sky analyses.

The primary-shift mean is

$$\left\langle \Delta\chi^2_{\rm primary} \right\rangle = 2.101464, \tag{262}$$

and the largest primary shift is

$$\max\left(\Delta\chi^2_{\rm primary}\right) = 3.669953. \tag{263}$$

The primary stacks are control-level; the CamSpec worst repeat 5.990520 lies in the declared watch/stress band while its primary best-of-N comparison remains control-level.

13.9 Combined-Stack Posterior Stability Diagnostic

The same fixed row is carried through Plik, full NPIPE CamSpec, and SPT/ACT posterior sampling with $A_{\rm lens} = 1$ and all gates fixed; all 6/6 jobs complete.

Stack	Median shift	Max final $R-1$	Classification
Plik TTTEEE + lowE + Planck lensing	−0.2238004	0.121641	Likelihood stable; convergence not claimed
NPIPE CamSpec + lowE + Planck lensing	−0.2238066	0.143463	Likelihood stable; convergence not claimed
SPT-3G TEEE + ACT DR6 lensing	+0.7375340	0.108998	Likelihood stable; convergence not claimed

Table 20: Locked combined-stack CMB posterior stability diagnostic. Full NPIPE CamSpec is used when CamSpec-lite is unavailable.

The median-shift mean and maximum are

$$\left\langle \Delta\chi^2_{\text{combined posterior}} \right\rangle = 0.0966423333, \qquad \max(\Delta\chi^2_{\text{combined posterior}}) = 0.737534. \tag{264}$$

The maximum reported final diagnostic is

$$\max(R-1) = 0.143463. \tag{265}$$

The shifts are control-level; with $R - 1 = 0.109$–0.143, this remains a stability diagnostic rather than a convergence claim [31].

13.10 Numerical Amplitude and Reionization-Depth Gate Closure

The final CMB row also supplies a direct arithmetic closure check of the fixed native-to-effective map,

$$\Omega_{c,\mathrm{eff}}h^2 = \Omega_{c,\mathrm{CPTG}}h^2\mathcal{D}_c, \tag{266}$$

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}}\frac{\pi}{3}, \tag{267}$$

$$\tau_{\mathrm{reio,eff}} = \tau_{\mathrm{reio,CPTG}}\mathcal{G}_T^2. \tag{268}$$

Using the frozen pre-map comparison inputs,

$$\Omega_{c,\mathrm{CPTG}}h^2 = 0.120885025627, \tag{269}$$

$$A_{s,\mathrm{CPTG}} = 2.04 \times 10^{-9}, \tag{270}$$

$$\tau_{\mathrm{reio,CPTG}} = 0.066, \tag{271}$$

the closure audit gives

$$0.120885025627\,\mathcal{D}_c = 0.1176858416205258, \tag{272}$$

$$2.04 \times 10^{-9}\left(\frac{\pi}{3}\right) = 2.136283004441059 \times 10^{-9}, \tag{273}$$

$$0.066\left(\mathcal{G}_T^2\right) = 0.05893087593426084. \tag{274}$$

The maximum relative closure difference is

$$0. \tag{275}$$

Thus the scalar-amplitude and reionization-depth maps close exactly; their geometric definitions are given in the preceding CMB construction section.

13.11 Coupled Gate Off-Lock Stress Sweep

A coupled off-lock sweep tests the gate relation while keeping the authoritative p_{ac} fixed; neighboring rows are diagnostic only, with n_s and N_{eff} held fixed. The sweep applies

$$\mathcal{G}_T(p_{\mathrm{ac}}) = \frac{p_{\mathrm{ac}}}{\pi}, \tag{276}$$

$$\mathcal{D}_c(p_{\mathrm{ac}}) = \sqrt{\mathcal{G}_T(p_{\mathrm{ac}})}\left[1 + \frac{1}{10\pi}\left(\frac{\pi}{3} - 1\right)\right], \tag{277}$$

$$\tau_{\mathrm{reio,eff}}(p_{\mathrm{ac}}) = \tau_{\mathrm{reio,CPTG}}\mathcal{G}_T(p_{\mathrm{ac}})^2. \tag{278}$$

The result is summarized in Table 21.

Rank	p_{ac} shift	Mean target-median shift	Max target-median shift	Status
1	+0.005	+2.212364	+4.574011	Control-level
2	0.000, locked	+2.297587	+4.677919	In basin
3	+0.010	+2.299781	+4.901645	In basin
4	−0.005	+2.584480	+5.078886	Mild stress
5	−0.010	+3.193148	+5.921761	Stress
6	+0.020	+3.543665	+7.271139	Stress
7	−0.020	+5.479216	+9.497230	Poor

Table 21: Coupled p_{ac}-gate off-lock stress sweep. The locked row remains inside the coupled-gate basin; neighboring rows are diagnostic stress evaluations only and do not retune p_{ac}.

The locked row ranks second and is only

$$0.085223 \tag{279}$$

above the lowest aggregate off-lock stress row; the difference is diagnostic only and does not select or modify p_{ac}.

13.12 CMB Implementation Boundary

The CMB implementation status is

$$\begin{gathered}\text{CPTG passes the fixed CMB comparison layer} \\ \text{through high-}\ell\text{ profile robustness and} \\ \text{combined-stack profile validation, with supporting} \\ \text{posterior stability and low-}\ell\text{/lensing diagnostics,} \\ \text{plus exact gate closure, with } A_{\mathrm{lens}} = 1.\end{gathered} \tag{280}$$

The fixed row requires neither optical-depth retuning nor phenomenological lensing enhancement; posterior convergence is not claimed where $R - 1 > 0.01$, and the gate arithmetic closes exactly.

14 DESI DR1 Adapter and Diagnostic Record

The DESI DR1 coordinate comparison is organized into three distinct layers. The DESI DR1 BAO measurements and full-shape analyses provide the published context for these coordinate-level diagnostics [7, 8, 9, 10]. The first layer is the compressed ShapeFit coordinate comparison using the official DESI DR1 HDF5 likelihood containers associated with the full-shape and BAO clustering products [11, 12]. The second layer is the BAO quarter-ruler coordinate-wrapper diagnostic. The third layer is a full-shape spectrum-shell AP/growth diagnostic using `desilike` coordinate behavior and the native HDF5 spectrum vectors.

The third layer is kept separate from the raw full-shape validation claim. It is a completed exploratory spectrum-shell diagnostic using the declared native HDF5 spectrum vectors and `desilike` coordinate behavior; a raw CPTG full-shape power-spectrum likelihood validation is not claimed here.

14.1 Compressed ShapeFit Coordinate Comparison

The ShapeFit HDF5 containers are checked with two reference vectors. The identity vector, where the model vector equals the observed compressed vector, gives

$$\chi^2_{\text{identity}} = 0. \tag{281}$$

The no-shift reference vector is

$$(q_{\text{iso}}, q_{\text{ap}}, d_m, d_f) = (1, 1, 0, 1), \tag{282}$$

which gives

$$\chi^2_{\text{no shift}} = 38.70438441. \tag{283}$$

The locked CPTG compressed-coordinate map follows the `desilike`/ShapeFit AP convention [12] and converts the transverse and radial coordinates into the ShapeFit basis by

$$q_{\text{iso}} = (q_\perp^2 q_\parallel)^{1/3}, \tag{284}$$

and

$$q_{\text{ap}} = \frac{q_\parallel}{q_\perp}. \tag{285}$$

The ShapeFit growth coordinate is used as a ratio to the fiducial growth rate,

$$d_f = \frac{f_{\text{CPTG}}}{f_{\text{fid}}}. \tag{286}$$

The tilt coordinate is held fixed at

$$d_m = 0. \tag{287}$$

This is an adapter choice for the present compressed test, not a derived CPTG prediction. A nonzero d_m should be introduced only after an explicit CPTG spectral-shape mapping is derived.

Thus the locked compressed ShapeFit vector is

$$\mathbf{x}_{\text{CPTG}} = \left((q_\perp^2 q_\parallel)^{1/3}, \frac{q_\parallel}{q_\perp}, 0, \frac{f_{\text{CPTG}}}{f_{\text{fid}}} \right). \tag{288}$$

For the DESI-like fiducial comparison, the direct HDF5 adapter gives

$$\chi^2_{\text{CPTG}} = 35.19040980558847, \tag{289}$$

with

$$\chi^2_\nu = 1.466267075232853, \tag{290}$$

and

$$\Delta\chi^2 = -3.51397460441153 \tag{291}$$

relative to the no-shift reference. For the Planck-like fiducial cross-check, the corresponding result is

$$\chi^2_{\text{CPTG}} = 38.46800119, \tag{292}$$

$$\chi^2_\nu = 1.60283338, \tag{293}$$

and

$$\Delta\chi^2 = -0.23638322. \tag{294}$$

This is a compressed coordinate-level pass. It is not a raw DESI power-spectrum likelihood validation.

14.2 BAO Quarter-Ruler Coordinate Wrapper

The BAO diagnostic uses the quarter-ruler transport relation

$$\mathcal{G}_T = \frac{p_{\rm ac}}{p_C}. \tag{295}$$

The corresponding ruler ratio is

$$\frac{r_{d,\rm comp}}{r_{d,\rm fid}} = \mathcal{G}_T^{-1/4} = 1.014261942163, \tag{296}$$

with the no-distance-shift AP compression scale

$$Q_\downarrow = \mathcal{G}_T^{1/4} = 0.985938600700. \tag{297}$$

At unity, the BAO compression wrapper gives

$$\chi^2_{\rm BAO}(q=1) = 20.12426210852749. \tag{298}$$

The official BAO wrapper exposes Ω_m, $\log\mathcal{L}$, and $\log\pi$ as runtime or pipeline parameters, but it does not expose $q_{\rm iso}$, $q_\parallel$, or $q_\perp$ as direct runtime inputs. Therefore the non-unity AP/ruler response is treated as a coordinate-wrapper diagnostic unless an upstream BAO theory calculator supplies the non-unity coordinates through the official likelihood path.

With the CPTG acoustic value of Ω_m and a free ruler ratio, the best non-boundary coordinate-wrapper result occurs near

$$\frac{r_{d,\rm comp}}{r_{d,\rm fid}} \simeq 1.014, \tag{299}$$

with

$$\chi^2 = 13.501774995995627. \tag{300}$$

The fixed quarter-ruler candidate gives

$$\chi^2_{\rm BAO}(\mathcal{G}_T^{-1/4}) = 13.505973446003. \tag{301}$$

An exponent scan using

$$r_d \mapsto \mathcal{G}_T^{\alpha} r_d \tag{302}$$

finds

$$\alpha_{\rm best} = -0.2435, \tag{303}$$

with

$$\chi^2_{\rm best} = 13.501376842627. \tag{304}$$

The CPTG quarter candidate is

$$\alpha_{\rm CPTG} = -\frac{1}{4}, \tag{305}$$

and is only

$$\Delta\chi^2 = 0.0045966 \tag{306}$$

above the grid best. The quarter-ruler relation is therefore a strong coordinate-wrapper diagnostic and remains an open independent derivation target. The near-quarter optimum is reported as a numerical correspondence rather than as a claim that DESI uniquely selected the exponent. It is not stated as a raw official non-unity q runtime likelihood response.

Layer	Quantity	Result	Classification
ShapeFit identity	χ^2	0	Native null check
ShapeFit no-shift	χ^2	38.70438441	Pipeline baseline
ShapeFit CPTG, DESI-like fiducial	χ^2	35.19040980558847	Coordinate-level pass
ShapeFit CPTG, Planck-like fiducial	χ^2	38.46800119	Cross-check
BAO unity wrapper	χ^2	20.12426210852749	Wrapper baseline
BAO ratio-scan optimum	χ^2	13.501774995995627	Free-ruler-ratio diagnostic
BAO exponent-scan optimum	χ^2	13.501376842627 at $\alpha = -0.2435$	Exponent-scan diagnostic
BAO quarter-ruler candidate	χ^2	13.505973446003	Supported diagnostic; derivation open

Table 22: DESI compressed-coordinate and BAO coordinate-wrapper status. The ratio-scan and exponent-scan optima are distinct pipeline outputs. The quarter-ruler row is a coordinate-wrapper diagnostic, not a raw official non-unity runtime response.

14.3 Full-Shape Spectrum AP/Growth Diagnostic

The full-shape spectrum-shell diagnostic evaluates whether the compressed-coordinate success extends into AP/growth spectrum coordinates. This diagnostic uses the native HDF5 spectrum vectors and `desilike` coordinate behavior. It does not inject a CPTG raw matter power spectrum, does not refit DESI nuisance parameters, and does not constitute a DESI full-shape validation.

The `desilike` AP definitions are [12]

$$q_\parallel = \frac{(D_H/r_d)}{(D_H/r_d)_{\rm fid}}, \tag{307}$$

and

$$q_\perp = \frac{(D_M/r_d)}{(D_M/r_d)_{\rm fid}}. \tag{308}$$

The associated AP and isotropic combinations are

$$q_{\rm ap} = \frac{q_\parallel}{q_\perp}, \tag{309}$$

and

$$q_{\rm iso} = q_\parallel^{\eta} q_\perp^{1-\eta}. \tag{310}$$

In isotropic mode,

$$q_\parallel = q_\perp = q_{\rm iso}. \tag{311}$$

The `BAOPowerSpectrumTemplate` default AP mode is the $q_\parallel, q_\perp$ convention. The HDF5 BAO products are mixed by tracer: BGS and QSO use $q_{\rm iso}$-type products, while ELG and LRG use $q_\parallel, q_\perp$-type products. The ShapeFit d_f coordinate scales the growth-rate ratio rather than inserting an absolute $f\sigma_8$.

The spectrum-shell diagnostics show that the locked quarter-down AP plus CPTG growth ratio improves the global spectrum-shell comparison, but the response is tracer- and multipole-dependent. The full sample improves by

$$\Delta(-2\log\mathcal{L}) = -184.04148385539065 \tag{312}$$

for the locked global quarter-down-plus-growth case. When split/tracer convention models are allowed, the best spectrum-shell diagnostic model is the BGS/ELG axis-repair plus LRG/QSO locked model, with

$$\Delta(-2\log\mathcal{L}) = -248.077305. \tag{313}$$

This means the spectrum-shell AP/growth layer is shaped by tracer response, multipole response, AP convention, and spectrum-window handling. It is not a clean universal full-shape validation layer.

Diagnostic channel	Observed response	Interpretation
BAO coordinate wrapper	Quarter ruler improves the compressed BAO coordinate comparison	Strong coordinate-wrapper ruler diagnostic
Full sample spectrum shell	Locked quarter-down AP plus CPTG d_f improves the global shell	Positive but not universal; split models do better
BGS monopole $\ell = 0$	$q_{\rm iso}$ supports quarter-down in low-, mid-, high-, and all-k bins	Ruler channel remains supported
BGS quadrupole $\ell = 2$	$q_{\rm iso}$ wants inverse/up in low-, mid-, high-, and all-k bins	Quadrupole AP/projection/window sensitivity
ELG transverse channel	$q_\perp$ wants up broadly, especially through $\ell = 2$ and combined blocks	Broad transverse tracer-response sensitivity
LRG/QSO	Locked quarter-down direction remains supported in low-, mid-, and all-k bins	Stable monopole/ruler-supported sector
Gross window/nuisance structure	BGS/QSO share $q_{\rm iso}$-type structure, while ELG/LRG share $q_\parallel, q_\perp$-type structure	Gross HDF5 structure alone does not explain the split

Table 23: DESI full-shape spectrum-shell diagnostics. These are comparison-layer diagnostics only, not full-shape validation claims.

The local k-bin and multipole diagnostics resolve the spectrum-shell response into tracer-, multipole-, and scale-dependent structure. The BGS monopole supports the CPTG quarter-ruler direction, while the BGS quadrupole reverses direction in every tested k-range, establishing that the reversal is not confined to high k. The ELG transverse $q_\perp$ response is likewise broad rather than isolated to high k, while LRG/QSO retain quarter-down support across the tested low-, mid-, and all-k sectors. The full-shape spectrum AP/growth result is retained as a completed exploratory spectrum-shell diagnostic with resolved tracer/multipole structure. This does not weaken the compressed ShapeFit coordinate pass or the BAO/monopole quarter-ruler support.

The DESI status is:

$$\begin{gathered}\text{compressed ShapeFit coordinate vector: passed,}\\ \text{BAO quarter-ruler coordinate wrapper: supported,}\\ \text{full-shape spectrum AP/growth: completed exploratory spectrum-shell diagnostic.}\end{gathered} \tag{314}$$

15 Late-Time Growth, Lensing, Distance, and Clock Verification Record

The fixed branch can be compared against compressed observational sectors using explicit coordinate conversions. External papers define observational anchors and data products; the χ^2, mask counts, residuals, and compressed pulls reported below are CPTG pipeline outputs rather than numbers supplied by those papers.

15.1 Compressed Weak-Lensing and CMB-Lensing S_8 Comparison

The late galaxy-lensing coordinate and the CMB-lensing coordinate are different compressed observables:

$$S_{8,\rm late}^{\rm CPTG} = 0.788071335232, \qquad S_{8,\rm CMBL}^{\rm CPTG} = 0.821191065949. \tag{315}$$

For published asymmetric galaxy-lensing uncertainties, this audit uses one rule throughout,

$$\sigma_{\rm sym} = \frac{\sigma_+ + \sigma_-}{2}. \tag{316}$$

Anchor	Published S_8	$\sigma_{\rm sym}$	CPTG residual
DES Y6 3×2pt	0.789 ± 0.012	0.0120	-0.077389σ
HSC-Y3 cosmic shear	$0.776^{+0.032}_{-0.033}$	0.0325	$+0.371426\sigma$
KiDS-1000 MetaCalibration	$0.789^{+0.020}_{-0.024}$	0.0220	-0.042212σ
KiDS-1000 lensfit reference	$0.776^{+0.029+0.002}_{-0.027-0.003}$	0.02812	$+0.429317\sigma$

Table 24: Compressed galaxy weak-lensing comparison rows used by the CPTG S_8 audit [26, 29, 27, 28]. For the lensfit row, statistical and quoted shear-calibration systematic uncertainties are combined in quadrature on each side before applying the common symmetric-error convention.

All four reported late-lensing comparison rows lie within 0.43σ of the transported CPTG coordinate. A purely descriptive inverse-variance summary of these four published analysis rows gives

$$S_{8,\rm WL}^{\rm desc} = 0.786438, \qquad \sigma_{\rm desc} = 0.009440, \qquad \frac{S_{8,\rm late}^{\rm CPTG} - S_{8,\rm WL}^{\rm desc}}{\sigma_{\rm desc}} \simeq 0.173025. \tag{317}$$

This is *not* a joint-survey likelihood or an independent combined measurement. The two KiDS rows are alternative analyses of the same underlying survey data, and the different surveys can have nontrivial covariance and modeling correlations. No combined statistical significance is assigned to this descriptive summary.

For CMB lensing, the appropriate published coordinate is

$$S_8^{\rm CMBL} \equiv \sigma_8 \left(\frac{\Omega_m}{0.3}\right)^{0.25}, \tag{318}$$

which matches the coordinate class used by the CPTG compressed CMB-lensing map. ACT DR6 reports 0.818 ± 0.022 from ACT lensing alone and 0.813 ± 0.018 from ACT DR6 combined with Planck NPIPE lensing [33].

CMB-lensing reference	Published $S_8^{\rm CMBL}$	CPTG residual
ACT DR6 lensing only	0.818 ± 0.022	$+0.14505\sigma$
ACT DR6 + Planck NPIPE lensing	0.813 ± 0.018	$+0.45506\sigma$

Table 25: CMB-lensing compressed-coordinate comparison for $S_{8,\rm CMBL}^{\rm CPTG} = 0.821191065949$ [33].

The ACT+Planck row contains overlapping lensing information with the ACT-only row, so the two are not inverse-variance combined. The CMB-lensing result is a compressed-coordinate diagnostic, not a full lensing likelihood validation.

15.2 Pantheon+ Supernova Distance-Shape Test

The Pantheon+ comparison uses the published statistical-plus-systematic covariance structure and analytically marginalizes one intercept. The explicit CPTG analysis mask used here retains 1624 non-calibrator rows; that row count is a pipeline output of the stated mask and is not presented as the published Pantheon+ sample size. The primary comparison therefore tests the shape of the supernova distance-redshift relation rather than an absolute H_0 calibration [5].

Model	χ^2	Reduced χ^2	$\Delta\chi^2$
Planck-like ΛCDM	1458.014422	0.898345	0
CPTG raw π branch	1458.058573	0.898373	+0.044151
CPTG acoustic projection	1458.644632	0.898734	+0.630210
DESI-like ΛCDM	1460.490455	0.899871	+2.476033

Table 26: Pantheon+ full-covariance distance-shape comparison. The CPTG mask retains 1624 non-calibrator rows and 1623 effective degrees of freedom after intercept marginalization.

The supernova sector therefore gives a distance-shape pass for the locked CPTG branch. It is not an H_0 calibration claim.

The exact mask definition and covariance handling are reproducibility inputs and should accompany the public analysis artifacts.

15.3 Age and Clock Consistency

The native and acoustic ages are the self-synchronized time coordinates of the native branch and its exact acoustic projection. They share the native phase $N_\pi = \ln a$, but each is integrated from its own rate history and the two times are not interchangeable.

For the two standard comparison rows, the Hubble normalization is held fixed at the transported CPTG CMB value,

$$H_{0,\mathrm{ac}}^{\mathrm{CPTG}} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{319}$$

and only the matter-fraction comparison coordinate is changed. For a flat ΛCDM comparison background,

$$t_0^{\mathrm{flat}}(\Omega_m; H_0) = \frac{2}{3H_0\sqrt{1-\Omega_m}}\,\sinh^{-1}\sqrt{\frac{1-\Omega_m}{\Omega_m}}. \tag{320}$$

Using $\Omega_m = 0.315$ for the Planck-like comparison coordinate gives $t_0 = 13.780328$ Gyr, while $\Omega_m = 0.300$ for the DESI-like comparison coordinate gives $t_0 = 13.970356$ Gyr. These rows are therefore internally generated comparison-coordinate ages under a common Hubble normalization, not quotations of the collaborations' full posterior age inferences.

Coordinate	t_0 Gyr	Planck-like difference [Gyr]	Planck-like ratio
CPTG raw π geometric	12.754121	−1.026208	0.925531
CPTG acoustic projection E_{ac}	13.885173	+0.104845	1.007608
CPTG CMB density-weighted acoustic diagnostic $E_{\mathrm{ac},\rho}$	13.926455176722689	+0.146127	1.010604
CPTG affine matched-normalization luminosity comparison	12.099622	−1.680706	0.878036
CPTG SH0ES-conditioned luminosity anchor scaffold	12.843057	−0.937271	0.931985
Planck-like coordinate ($\Omega_m = 0.315$)	13.780328	0	1
DESI-like coordinate ($\Omega_m = 0.300$)	13.970356	+0.190028	1.013790

Table 27: Present branch-clock comparison across the CPTG native, exact acoustic, affine matched-normalization luminosity, SH0ES-conditioned luminosity-anchor, and acoustic-density diagnostic coordinates, together with flat-ΛCDM comparison coordinates. The affine luminosity row uses $H_L^{\mathrm{aff}}(a) = \sqrt{10/9}\,H_\pi(a)$; the anchor row retains the separate historical $73.17\,E_{\mathrm{ac},\rho}(a)$ scaffold. These branch-clock coordinates are not competing estimates of one universal physical age.

The CPTG acoustic projection lies within about one percent of the Planck-like comparison age. The CMB density-weighted value is the diagnostic defined in Section 11.1; it is not a separate acoustic branch or a second acoustic clock. The affine luminosity value is the comparison-clock coordinate implied by the exact constant normalization in Eq. (184), whereas the 12.843057 Gyr value is the distinct SH0ES-conditioned scaffold of Eq. (196). Their substantial clock difference reflects different full histories despite nearly identical present endpoints. Neither luminosity row is a claim for a second physical age of the Universe or an independently initialized cosmology.

16 Implementation Status and Scope Boundaries

The status matrix below is controlling. CMB validation refers to the fixed-row profile record with $A_{\mathrm{lens}} = 1$ and fixed gates; low-ℓ, lensing, and MCMC results remain supporting diagnostics, with no convergence claim where $R - 1 > 0.01$ [31].

The fixed-row robustness categories are

$$\Delta\chi^2 \le 5 \quad \text{(control)}, \qquad 5 < \Delta\chi^2 \le 10 \quad \text{(watch/stress)}, \qquad \Delta\chi^2 > 10 \quad \text{(decisive stress)}, \tag{321}$$

and are pragmatic likelihood-robustness categories rather than universal confidence limits [3, 37].

Sector failure conditions remain those registered upstream: no post-target retuning may rescue a failed primordial, direct-reaction, or first-order network gate; compressed galaxy- and CMB-lensing coordinates remain distinct; and DESI compressed-coordinate status is separate from any raw nuisance-preserving full-shape validation.

16.1 Sector Status Matrix

Each sector compares a fixed CPTG object to the observational coordinate, likelihood summary, or compressed variable appropriate to that sector. Status labels are intentionally matched to the strength of the implemented test.

Table 28: CPTG comparison status by sector.

Sector	CPTG comparison object	Status
Cosmological branch architecture	$\mathfrak{B}_\pi \to \{\mathfrak{B}_{\rm ac}, \mathfrak{B}_{\rm lum}\}$	One native branch; acoustic and luminosity branches are projections
Bound-system/native-phase interface	$a_\star, R_c, a_\pi, R_C, \eta_\star, H_{\rm struct}, \mathcal{K}_\star$	Cross-scale acceleration/radius/rate identities defined; fixed-local-state, native-locking, and physical-evolution trajectories kept distinct
Background expansion	$E_\pi(a)^2$, $E_{\rm ac}(a)^2$, $H_L^{\rm aff}(a)$, $H_{\rm lum}^{\rm anchor}(a)$	Native background and exact acoustic projector defined; affine luminosity comparison has exact native normalized shape; SH0ES anchor retained separately
Age and clock map	$t_\pi(a)$, $t_{\rm ac}(a)$, $t_L^{\rm aff}(a)$, $t_{\rm lum}^{\rm anchor}(a)$	Branch-specific comparison-clock integrals defined at common phase; affine and anchor luminosity clocks kept distinct
Luminosity comparison	$\mathsf{C}_{\Pi,\star} = I/9$, $C_H/C_R = 1/9$, $\mathcal{R}_3 = \Omega_{m,\rm ac}/\mathcal{G}_T = 1/3$, $\mathcal{G}_L^{\rm aff} = 1+\mathcal{R}_3^2 = 10/9$, $E_L^{\rm aff} = E_\pi$	Matched polarization, structural-coefficient, and acoustic-rank derivations close on the same 1/9 increment; affine comparison prescription then registers 10/9; SH0ES supplies independent external validation; full covariant stability and unique microscopic nonmatched projector not claimed
Acoustic–luminosity midpoint	$\omega_{\rm AL} = 1/2$, $\mathcal{M}_{\rm AL}^{\rm aff}(a)$, $H_{0,\rm mid}^{\rm aff}$	Symmetric midpoint prescription applied primarily to theory-constructed acoustic and affine luminosity coordinates; historical anchor midpoint retained separately; no third branch
Cosmological horizon	$\chi_\star$, $L_\star/R_C$, $z_{\rm enter}$	Common-phase horizon coordinates registered; $L_\star/R_C = 0.53792156 < 1$ at last scattering
Supernovae	Standardized distance residuals with marginalized intercept	Full-covariance shape pass
BAO	D_M/r_d, D_H/r_d, D_V/r_d, AP coordinates	Coordinate comparison supported; externally fitted comparison-layer absolute normalization declared; no native absolute-ruler derivation claimed
Quarter-ruler	$\mathcal{G}_T^{-1/4}$	Supported wrapper diagnostic; independent derivation open
BBN background	$\Theta_{\rm BBN}^{\rm CPTG} = (\eta_{10}^{\rm BBN}, N_{\rm eff}^{\rm CPTG}, \tau_7)$, with background read from the gate-off native row	PRIMAT v0.3.2 large-network full-coordinate validation pass; D/H and Y_p within preregistered controls
Nuclear reaction observable	$S_{12}^{\rm CPTG}(0) = K_{S0} I_{C7}$, $S_{12}^{\rm CPTG}(E) = S_{12}^{\rm CPTG}(0) F_{\rm ext}(E)$, $\mathcal{R}_{pD}^{\rm CPTG}(T)$	Commissioned $D(p,\gamma)^3$He projection; 13/13 Mossa/LUNA S rows and 17/17 rate rows inside published intervals; external shape carrier declared
Nuclear reaction response	$\mathcal{C}_{r,\mathcal{N}}^{(1)} = \mathsf{P}_3 \mathsf{L}_{\mathcal{N}}^{(1)} \mathcal{A}_{r,\mathcal{N}}^{(1)} \mathbf{s}_r$	Network-conditioned first-order pushforward; PRIMAT 48/48 primary resolved tests pass, PArthENoPE supplies independent Reaction-20 transfer
Mass-seven abundance transport	$\tau_7 = \ln \pi$, $\mathcal{S}_7 = 1/\pi$, live ^{7}Li/^{7}Be sink	23/23 scientific gates pass; measured survival 0.318366498376680; lithium endpoint downstream diagnostic
Galaxy weak lensing	$S_{8,\rm late}^{\rm CPTG} = \mathcal{G}_T S_{8,\rm anchor}$	DES Y6/HSC/KiDS compressed support; maximum reported $\lvert\text{pull}\rvert = 0.429317\sigma$
CMB lensing	$S_{8,\rm CMBL}^{\rm CPTG}$	ACT DR6 compressed-coordinate diagnostic support
CMB profiles	Locked acoustic-transport row	Profile-validation pass; exact gate closure

Continued on next page

Sector	CPTG comparison object	Status
CMB posterior sampling	Same locked row	Likelihood-stability diagnostic; conventional convergence not claimed
CMB curvature-transport spectrum bridge	$C_T = \Phi_\pi/\Phi_0$, $\mathcal{R}_T = P_\pi/P_0 = C_T^2$	Route B Option 1 bridge locked; native spectral-response implementation not claimed
CMB SkyPi current map	Fixed $C_\ell^{\mathrm{SkyPi,src}} = q_\ell^2 C_\ell^{\pi,\mathrm{src}}$; observed-phase forward map	SkyPi current source/mask-response convention [42]; legacy fitted map-closure test separately preserved [44]
DESI ShapeFit	Compressed ShapeFit vector	Coordinate-level pass
DESI full-shape spectrum	AP/growth spectrum-shell response	Completed exploratory diagnostic; raw nuisance-preserving full-shape validation not claimed

Each comparison layer therefore has a fixed object and a documented operational failure condition. A distance-sector implementation fails if the fixed distance relation misses covariance-level supernova or BAO structure after the appropriate coordinate map is applied. The primordial-network implementation fails if the prospectively frozen baryon or radiation coordinate cannot be reproduced natively, if the live mass-seven response misses the preregistered survival tolerance, or if the gate disturbs D/H, Y_p, or ^{3}He/H beyond the frozen controls. The observed lithium endpoint remains downstream of that decision. Growth/lensing status is assessed against the appropriate compressed coordinate rather than by mixing galaxy-shear and CMB-lensing definitions.

For DESI, compressed-coordinate failure and raw full-shape failure remain separate: the former concerns the locked ShapeFit/BAO mappings; the latter requires the full nuisance-preserving AP, RSD, tracer-window, bias, counterterm, stochastic, and covariance machinery.

17 Limitations and Open Closure Conditions

Open boundaries are limited to the following. The spectrum bridge does not yet supply a native scale-to-spectrum law for $C_T(a,k)$, and its scalar metric/Poisson/Weyl relations remain external translation coordinates. The BAO quarter-ruler still lacks an independent geometric derivation. The luminosity completion has $W_\star = p_{\mathrm{pol}}$, $\mathfrak{K}_\star = 1/2$, $\mathsf{C}_{\Pi,\star} = I/9$, $C_H/C_R = 1/9$, and $\mathcal{R}_3 = \Omega_{m,\mathrm{ac}}/\mathcal{G}_T = 1/3$ with $\mathcal{R}_3^2 = 1/9$, giving $\mathcal{G}_L^{\mathrm{aff}} = 10/9$ and $E_L^{\mathrm{aff}} = E_\pi$; however, complete covariant stabilization, unique dynamical selection of U^μ, and a unique microscopic luminosity operator or universal metric rescaling are not claimed. The SH0ES-conditioned scaffold in Eqs. (196)–(199) remains observational only, and $E_{\mathrm{ac},\rho}$ remains a diagnostic overlay. DESI full-shape status remains exploratory, $d_m = 0$ remains a fixed ShapeFit adapter setting, CMB posterior runs remain likelihood-stability diagnostics rather than conventional convergence claims, and no universal phase-evolution law is imposed on object-dependent $a_\star$. The midpoint retains $\omega_{\mathrm{AL}} = 1/2$ and defines neither a third branch nor an independently fitted Hubble parameter.

These limitations leave the already stated fixed-row CMB, Pantheon+, compressed-lensing, PRIMAT full-coordinate, and first-order pushforward results within their declared scopes.

18 Likelihood, Software, and Evidence Provenance

The provenance matrix separates external data/method sources from CPTG-generated evidence and binds the major claim lanes explicitly.

Table 29: Paper-wide provenance matrix. Artifact names identify the evidence lane; cryptographic identities for sealed bound-system and primary nuclear authority lanes are given in the Data, Code, and Evidence Availability section.

Claim lane	Upstream data / authority / artifact	Freeze or conditioning rule	Claim boundary
Native pi branch	CPTG framework and public computational materials [38, 51]	Native branch fixed before sector comparisons	Native/derived geometric quantities only
Bound-system/native-phase and structural-observation interface	NMBS-C7447 native-locking, local-invariance, and direct comparison analyses; Structural Mode N/CSMI authority; XMM-VID1-2075 resolved-inference measurement test; Dragonfly 44 resolved tensor and higher-moment analysis [45, 46, 47, 49, 48, 50]	Native phase is selected independently of the bound-system solve; each comparison trajectory, structural readout, and observation operator is declared before cross-scale or higher-order interpretation	Defines the cross-scale identities, Structural Mode readout, and observation-operator hierarchy; does not establish a universal phase-evolution law for $a_\star$ or permit higher-order residuals to retune the retained field
Luminosity affine comparison	Parent CPTG polarization homogeneity [38]; matched carrier response; structural coefficients C_H, C_R; acoustic quantities $\Omega_{m,\mathrm{ac}}, \mathcal{G}_T$; affine registration specified in Section 11.2	No SH0ES quantity enters the matched-carrier, structural-ratio, or acoustic-rank derivation of the 1/9 increment; SH0ES is applied afterward as independent external validation, while the historical anchor history is retained separately	Three algebraically distinct CPTG routes give 1/9, and the registered affine map gives 10/9; stationary-point result does not constitute a complete stability theorem; no universal physical-metric rescaling or unique microscopic nonmatched projector claimed
Acoustic–luminosity midpoint	Registered weight $\omega_{\mathrm{AL}} = 1/2$ applied primarily to the acoustic and affine matched luminosity endpoints	Endpoint quantities are evaluated at the same N_π before midpoint formation; historical SH0ES anchor midpoint retained separately	Symmetric comparison prescription; no independent midpoint fit and no third branch
CMB fixed-row likelihood tests	Planck, NPIPE/CamSpec, SPT-3G, ACT likelihood/data products and Cobaya machinery [2, 3, 37, 35, 36, 33, 34, 30, 32, 31]	CPTG row and gates fixed before each likelihood evaluation	Profile validation plus stated stability diagnostics; conventional MCMC convergence not claimed where $R - 1$ is too large
Route B spectrum bridge	CPTG Route B evidence chain [41, 51]	Squared-amplitude response mode fixed before bridge stress tests	Bridge validation only; no completed native scale-to-spectrum transport law
CMB SkyPi current comparison	SkyPi frozen source and selected Planck response [42]	Pi inputs, PRIMAT helium, and low-ℓ power factors frozen; measured harmonics supply phases	Current unfitted map/spectrum comparison; legacy fitted map-closure study retained separately [44]
Primordial $A = 7$ transport	PRIMAT v0.3.2 primary authority [14]; $A = 7$ authority lock and companion lithium analysis [23]; full evidence identity in Section 20	Preregistration and radiation mapping frozen before target abundance exposure; 23 scientific gates fixed in advance	Tested PRIMAT large-network configuration only; lithium endpoint downstream
Direct $D(p,\gamma)^3\mathrm{He}$ observable projection	CPTG commissioning paper, Solar Fusion II, Mossa/LUNA and comparison datasets [15, 16, 22, 17, 18, 19, 20, 21]	Zero-energy CPTG normalization fixed; finite-energy/thermal external carrier declared	Validates the commissioned observable projection; external energy shape is not native CPTG geometry
First-order nuclear reaction pushforward	PRIMAT paper-ready evidence, accepted six-anchor source/Jacobian results, and PArthENoPE Reaction-20 authority [22, 24]; full filenames and checksums in Section 20	Source/Jacobian predictions frozen before authoritative endpoint coefficients are opened	Network-conditioned pushforward; no universal code-independent endpoint matrix
Pantheon+ distance shape	Pantheon+ data and covariance [5]	CPTG distance projection fixed; one intercept marginalized by declared protocol	Full-covariance distance-shape claim, not an H_0 calibration
Weak/CMB lensing compressed coordinates	DES Y6, HSC-Y3, KiDS-1000, ACT DR6, Planck anchor [26, 29, 27, 28, 33, 2]	Named anchor/coordinate fixed before residual calculation; overlapping analyses not combined as independent measurements	Compressed-coordinate support only
DESI ShapeFit / BAO / spectrum shell	DESI DR1 products and `desilike` conventions [7, 8, 9, 10, 11, 12]	Adapter choices and frozen ruler inputs declared before each comparison	ShapeFit coordinate pass; BAO coordinate comparison supported with externally fitted comparison-layer absolute normalization declared; quarter-ruler strongly supported as a diagnostic with independent derivation open; full-shape spectrum-shell retained as an exploratory diagnostic

Posterior sections retain their reported $R-1$ values and the Cobaya 0.01 convergence boundary [31].

CPTG-generated outputs remain bound to the evidence identities below where available; unsealed outputs remain pipeline results [51].

19 Reference Summary and Usage Boundaries

This guide fixes the common language of the CPTG geometric π branch: one native branch with acoustic and luminosity projections sharing $N_\pi = \ln a$ and branch-specific clocks $H_b dt_b = dN_\pi$. The luminosity normalization is multiply closed by $\mathsf{C}_{\Pi,\star} = I/9$, $C_H/C_R = 1/9$, and $\mathcal{R}_3 = \Omega_{m,\mathrm{ac}}/\mathcal{G}_T = 1/3$, so $\mathcal{G}_L^{\mathrm{aff}} = 10/9$, $H_L^{\mathrm{aff}} = \sqrt{10/9}\, H_\pi$, and $E_L^{\mathrm{aff}} = E_\pi$; SH0ES is an external validation, not the defining input. The parameter registry is the primary lookup authority, while the sector sections state the corresponding constructions, scope boundaries, and evidence status. Structural Mode, nuclear-response, CMB, distance, lensing, BAO, ShapeFit, and network details are developed further in their dedicated papers and evidence releases. Cross-sector agreement is not presented as proof of CPTG; this reference standardizes definitions, dependency order, freeze rules, and permissible interpretation.

20 Data, Code, and Evidence Availability

External data and software are cited at their source. For sealed CPTG evidence lanes, the archive filename plus SHA-256 digest defines the immutable identity; repository locations are distribution mirrors [51].

20.1 Bound-System Native-Phase and Resolved-Inference Evidence

The bound-system/native-phase interface is supported by the two NMBS-C7447 comparison trajectories and by the XMM-VID1-2075 resolved-inference measurement test. These archives document distinct questions and are not interchangeable evidence authorities.

NMBS-C7447 native-locking evidence

`CPTG_NMBS-C7447_NativeLock_Structural_Evidence_20260904_r04_FINAL_EVIDENCE.zip`

`SHA-256: fc3076d3ee5a83fd17f1a1a75fe73844bb36974440a0a204d7009c446247420a`

NMBS-C7447 local-structural-invariance evidence

`CPTG_NMBS-C7447_LocalStructuralInvariance_Evidence_20260905_r02_RECONSTRUCTED_FINAL_EVIDENCE.zip`

`SHA-256: aa729c102ddecf768d3e06f6a4a39cf27ebbc846abe9a1470c5ed3507d5a1234`

The r02 archive is the reconstructed evidence authority for the local-structural-invariance lane. It is a reconstructed artifact and is not represented as a byte-for-byte recovery of the unavailable r01 archive.

XMM-VID1-2075 full-Jeans structural-inference evidence

`XMM-VID1-2075_FullJeans_aStar_Inversion_Test_20260904.zip`

`SHA-256: d6571f12fd815ef94fa70891a494c4ae9b87dfcb07145c1dbb7750d414887455`

The NMBS archives preserve the two deliberately different native-phase continuation tests from the same observed state. The XMM archive preserves the resolved Jeans inversion used to test how the inferred $a_\star$ and its downstream cross-scale ratios depend on the measurement interface. Together they support the reference-level distinction between phase selection, phase continuation, and structural inference [47, 48].

20.2 PRIMAT Full-Coordinate Primordial-Network Evidence

The PRIMAT baryon, radiation, and mass-seven full-coordinate validation is bound by the following authority lock.

Authority lock

`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_AuthorityLock_20260819_2359EDT_r01.zip`

SHA-256

`43b65036e229dc5f04d445f5890399236aaf546c598f1711e572043db6b05bec`

Recognized final-evidence archive

`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_20260819_2344EDT_r05_FINAL_EVIDENCE.zip`

SHA-256

`9c44543cd2497e72170c0a64173423c9b3bf2f1f3924c9d4e77c16a318fb8b4c`

The authority lock binds the full-coordinate evidence lane to the recognized final-evidence archive and preserves the preregistered baryon/radiation mapping, native-network execution, mass-seven transport result, scientific gates, and associated audit record.

20.3 PRIMAT First-Order Reaction-Response Evidence

The PRIMAT first-order network-conditioned reaction-response validation is preserved separately from the full-coordinate mass-seven evidence.

Primary validation archive

`CPTG_PRIMAT_PaperReadyValidationEvidence_20260817_r02.zip`

SHA-256

`b41c8ee49477d8330deb56c85a230d8dcc1d4bba80a1405251a4d6ea7f9b3205`

Prediction-freeze digest recorded inside the primary validation archive

`679e3e6fc7432c0592004b9bf0985b0756323ceb327050f3303ca6c816f31bff`

This digest identifies the frozen first-order prediction layer recorded by the paper-ready authority archive; it is not presented as a separately distributed standalone file.

Accepted six-anchor source/Jacobian results

`CPTG_PRIMAT_SIXANCHOR_NATIVE_SOURCE_JACOBIAN_RESULTS_20260817-202839.zip`

SHA-256

`8511f3f79d4298d077a32625859440e01f3b530a3eda6a3e999582e1f938baf8`

These identities bind the frozen source/Jacobian predictions to the accepted six-anchor native endpoint-response evaluation and preserve the separation between prediction construction and authoritative endpoint-reference exposure.

20.4 PArthENoPE Second-Authority Transfer Evidence

The independent PArthENoPE Reaction-20 transfer used as the second numerical authority within its declared shared-endpoint scope is preserved in the following authority bundle.

PArthENoPE authority bundle

`CPTG_v129_r109_PArthENoPE_CPTG_Reaction20_ReplicationAuthorityBundle_20260731_r02.zip`

SHA-256

`3d42d1cb1d710248841db8d7b1ceafcc7569f092b9b35c9f4114ce9615074cfc`

This archive preserves the independent network-transfer evidence cited for the PArthENoPE comparison and does not broaden that result beyond the Reaction-20 transfer scope defined in the text.

20.5 CMB, Supernova, Lensing, and DESI Evidence

The CMB, Pantheon+, weak-lensing, CMB-lensing, ShapeFit, BAO, and DESI spectrum-shell results are supported by the cited observational data products, likelihood implementations, CPTG-generated masks, scan tables, likelihood outputs, residual calculations, and numerical summaries described in their respective sections.

These quantities remain classified as *pipeline results* because they are outputs of the declared comparison and likelihood procedures rather than native CPTG invariants or frozen theory parameters. Where a separate cryptographically sealed release is available, its checksum identity may additionally be used to verify the exact distributed artifact. The presence or absence of a separate archival checksum does not change the evidentiary status of the reported calculation or its stated claim boundary.

20.6 Evidence Identity and Distribution

For cryptographically sealed evidence lanes, reproducibility is anchored by three elements:

1. the named evidence archive or result object;
2. its SHA-256 digest; and
3. the corresponding calculation, freeze rule, or validation scope stated in this article.

The public CPTG repository and its releases index provide the distribution entry points for the supporting materials [51]: `https://github.com/CLG2025/CPTG`

Repository and release paths are distribution mechanisms rather than scientific identities. Where a checksum-bound archive is listed above, the archive filename and SHA-256 digest provide the immutable reference for independently obtained copies.

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